# Scaling Multi-Task Learning Beyond Homogeneous Tasks: Loss Dynamics and Gate Selection in 12-Task Financial Recommendation

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**Abstract.** Multi-task learning (MTL) architectures such as MMoE and PLE have been validated almost exclusively on 2–4 homogeneous tasks (e.g., CTR + CVR). We report empirical findings from scaling PLE to **13 heterogeneous tasks** — 7 binary, 3 multiclass, 3 regression — in a financial product recommendation system with 7 structurally distinct experts and 1M synthetic customers (the final v14 benchmark uses 12 of these — 7 binary, 2 multiclass, 3 regression — after a leakage-motivated removal of segment_prediction; each finding's task composition follows the run it reports). Fourteen findings organize around one thread — **what silently fails when MTL is pushed past the homogeneous-task regime, and the measurement discipline needed to catch it** — across four themes. **Loss dynamics and gating** (Findings 1–6): an uncertainty-weighting implementation bug silently suppresses minority-type tasks (+0.018 NDCG@3 when fixed); softmax gating outperforms sigmoid for heterogeneous task mixes, **reversing** the homogeneous-setting advantage; learned uncertainty weights converge identically across architectures; shared-bottom baselines overfit extended epoch budgets that gated PLE variants absorb without regularisation; GroupTaskExpert pre-gating degrades multiclass performance in mixed-type groups; gate entropy analysis shows extraction-layer specialization (entropy ratios 0.33–0.88) while attention-level aggregation collapses to uniform averaging (ratio 1.00), and composite val-loss is an unreliable checkpoint signal once regression tasks are present. **Fusion augmentation trade-offs** (Finding 7): a 9-way comparison isolates two positive recipes on disjoint axes — output-space boosting with gradient isolation (BRP-detached, hard-task rescue) and inverse-gate auxiliary supervision (NEAS, aggregate-AUC gain) — that do **not** compose additively. **Causal expert reinterpretation** (Findings 8–13): the causal expert's adjacency matrix $\boldsymbol{W}$ collapses to zero in every trained checkpoint examined, rendering its forward equivalent to a plain MLP; a two-part patch (NOTEARS reconstruction loss + initialisation rescale) restores DAG learning at zero task-metric cost; routing the functional DAG into consumable outputs yields a Causal Explainability Head (CEH, per-sample attribution, Pearl Rung 1) and a Causal Guardrail (CG, z-space-Mahalanobis OOD detection at 100% TPR / 5% FPR on three synthetic probes); W-amplification via init $0.1 \rightarrow 0.3 + \lambda_{\text{recon}} 0.5 \rightarrow 2.0$ grows the adjacency matrix 14 × in Frobenius norm with zero primary-task cost, establishing that the "decorative DAG" is a training-choice artefact, not an architectural constraint; and a counterfactual probe (CCP, Pearl Rung 3) shows that at baseline W scale the DAG carries 0.16% of the counterfactual effect but at the amplified scale the mediation ratio rises to **32%** at median and **61%** at the 95th percentile — a preliminary signal, which we flag as future work rather than a validated result, that Pearl's Rung 3 may be viable on the amplified teacher; and a CEH v3 variant replacing the causal-encoder-output target with a specific task-logit gradient target is an honest negative result — the head re-collapses to a global importance pattern despite training signal, showing that target design for attribution is more sensitive than the v1→v2 pivot suggested. **Serving-side distillation control** (Finding 14): distilling the 12-task teacher into per-task gradient-boosted students passes operational gates (AUC gap $\leq 0.0125$, student calibration error $\leq 0.0114$) while behavioural similarity shows a structural floor (teacher–student agreement 0.75–0.82, ranking correlation 0.78–0.91 across all seven distilled tasks) that is invariant across three generations of gate semantics scored on identical predictions; a single-tier gate over both metric families therefore always fails — and an always-failing gate invites bypass — so the control splits into a blocking operational tier and an informational behavioural-diagnostic tier. We distill these observations into practical guidelines for practitioners scaling MTL beyond the homogeneous-task regime.

**Keywords:** Multi-task learning, Progressive Layered Extraction, Mixture of Experts, gate selection, uncertainty weighting, loss dynamics, heterogeneous tasks

# 1 Introduction

Multi-task learning (MTL) promises parameter efficiency and positive transfer by jointly optimizing related tasks [1]. In recommendation systems, MTL has become standard practice: MMoE [2] introduced multi-gate mixture-of-experts to handle task conflicts, PLE [3] added progressive extraction layers with shared and task-specific expert separation, and AdaTT [4] enabled adaptive inter-task transfer.

However, virtually all published MTL architectures for recommendation are validated on **2–4 homogeneous tasks** — typically click-through rate (CTR) and conversion rate (CVR), both binary classification problems with aligned gradient directions. When we attempted to scale PLE to **13 heterogeneous tasks** — 7 binary, 3 multiclass (4 to 50 classes), and 3 regression — several assumptions broke down in ways that existing literature does not address.

The 13-task configuration was not a design preference but a constraint. Financial regulations mandate distinct prediction targets: suitability assessment, fairness monitoring across protected attributes, churn early-warning, and product-level acquisition propensity each require a separate supervised signal. In particular, the Korean FSC Financial Sector AI Guidelines [5], in force since 22 June 2026, recommend global and local explainability (reliability principle, §4.4) and fairness monitoring across protected attributes, grounding these distinct prediction targets in an enacted norm (the guideline is self-applied and does not mandate any specific task configuration). Meanwhile, limited infrastructure — a single desktop GPU (12GB VRAM) and a 3-person team — precludes maintaining separate models per task. The result is a regime that large-scale CTR teams have no reason to enter (they can afford model-per-task) but that resource-constrained regulated industries are forced into.

This paper reports fourteen empirical findings from this scaling experience, organised into four themes: loss dynamics and gating (Findings 1–6), fusion augmentation trade-offs (Finding 7), causal expert reinterpretation (Findings 8–13), and serving-side distillation control (Finding 14). We make no claims of state-of-the-art performance; instead, we document **phenomena and practical guidelines** that emerge when MTL is pushed beyond the homogeneous-task regime.

**Position relative to companion papers.** This work sits between two companion papers that share the same 12-task benchmark, the same 7-expert PLE backbone, and the same v14 phase0 data pipeline.[1] Its own thread is narrow and deliberate: **what silently fails when MTL is pushed past the homogeneous-task regime, and the measurement discipline that catches it.** Paper 1 (Heterogeneous Expert PLE: An Explainable Multi-Task Architecture for Financial Product Recommendation)[2] presents the architecture and a system-level summary of the principal ablation findings; the present paper is the **detailed empirical companion** to that summary, reporting (a) full hypothesis discrimination across A–E (signal cleaning, epoch budget, ensemble, gate overfitting, regularisation) and per-task v14 numbers across all 12 tasks (Findings 1–6) — where these coincide with Paper 1 we cite its figures rather than re-deriving them — (b) a 9-way fusion- mechanism comparison that isolates the design space of "what kinds of fusion-augmentation work on top of CGC" (Finding 7), (c) a causal expert reinterpretation arc that has no counterpart in Paper 1 (Findings 8–13), and (d) a measurement-side analysis of the serving stack's distillation fidelity gate (Finding 14) whose governance counterpart — audit-trail semantics and operator-override risk — is treated in Paper 2′s discussion of fidelity-gate semantics. Paper 2 (From Prediction to Persuasion: Agentic Recommendation Reason Generation for Regulatory-Compliant Financial AI)[3] is self-contained: its GDPR Art. 22 / EU AI Act Art. 13 argument is carried by the distilled student's SHAP attributions, a three-agent reason pipeline, and an HMAC-signed hash-chained audit log, and does not depend on any output of this paper. Two of our findings can, however, be offered to that audit log as an **optional forensic enhancement** — the per-sample feature-attribution vector produced by the Causal Explainability Head (Finding 9) and the per-prediction reliability score produced by the Causal Guardrail (Findings 10–11), which Paper 2 can route into `log_attribution` / `log_guardrail` records. We are explicit that this is an enhancement, not a dependency: the per-prediction causal audit runs on the PLE teacher rather than the served student, so wiring it into production traffic — and the W-amplification (Finding 11) that would sharpen the reliability signal — is future work, not a delivered capability.

Our contributions:

- A diagnosis of how Kendall et al.'s uncertainty weighting silently fails when per-task loss weights are omitted (Section 4.1).
- Evidence that **gate type selection depends on task-type homogeneity**, not on architectural sophistication (Section 4.2).
- A demonstration that uncertainty weights converge identically across architectures, limiting their protective role (Section 4.3).

- Analysis of epoch budget sensitivity in structure comparison (Section 4.4).
- A cautionary finding on pre-gating task grouping (GTE) with mixed-type groups (Section 4.5).
- Gate entropy analysis showing that CGC extraction-layer specialization is real and task-dependent, while attention-level aggregation collapses to uniform averaging; and a demonstration that val-loss is a misleading checkpoint criterion when regression and classification tasks coexist (Section 4.6).
- A 9-way comparison of fusion augmentations on top of the CGC baseline that identifies **two positive recipes on disjoint axes** and shows that they are **not additive**. Five representation-additive fusions (loss-level adaTT, AdaTT-sp, complementary-gate recovery, uncertainty-conditioned expert bank, and MV BRP) all degrade AUC monotonically with intervention invasiveness. **Output-space boosting with shared-expert gradient isolation** (BRP-detached) ties CGC on aggregate AUC ($\Delta = -0.0007$; best epoch exceeds baseline by $+0.0008$) while lifting F1 macro $+0.007$ and NDCG@3 $+0.015$ and retaining $+256\%$ relative rescue on the hardest multiclass task. **Training-time load-balancing regularisation** (NEAS — auxiliary supervision on the inverse-gate aggregation) is the first mechanism of the family to actually raise aggregate AUC ($\Delta = +0.0011$), with a monotone-increasing trajectory and near- uniform per-task lifts. Stacking the two positive recipes collapses NEAS's AUC gain because the shared experts cannot simultaneously be generalists (NEAS) and primary-supporting specialists (BRP-detached). The guidance is per-objective: NEAS for aggregate AUC and cross-task robustness, BRP-detached for hard-task rescue, do not stack (Section 4.7).
- A structural diagnostic showing that the causal expert's learnable adjacency matrix $\boldsymbol{W}$ collapsed to zero across every trained checkpoint we examined (four local, two upstream on-prem), rendering the expert's forward pass equivalent to a plain MLP in spite of its NOTEARS regularisation. The failure is a saddle-point problem (both task-loss and reconstruction gradients have a $\boldsymbol{W}$ factor that vanishes at the initialisation scale) and resolves under a two-part patch: add the original-paper reconstruction term ($\|\boldsymbol{z} - \boldsymbol{z}\boldsymbol{W}^2\|_F^2$) and rescale initialisation from 0.01 to 0.1. Post-patch the expert learns a valid sparse DAG ($\boldsymbol{W}$ Frobenius 0.34, 7.3% edges active, $h(\boldsymbol{W}) = 0$), but aggregate task metrics are unchanged — the DAG exists structurally but is not routed into prediction by the current architecture (Section 4.8).
- Two Axis-3 candidates that **route** the now-functional DAG into consumable outputs:

  Causal Explainability Head (CEH), a small MLP mapping the causal expert's output to a per-sample per-feature attribution vector trained via MSE against gradient $\times$ input. MV result preserves the primary AUC within noise ($+0.0015$ over the post-patch softmax baseline), marginally strengthens the DAG itself ($\boldsymbol{W}$ Frobenius $0.338 \to 0.366$, sparse edges $7.3\% \to 8.5\%$), and produces a live attribution vector for downstream audit-log persistence. A post-hoc quality evaluation revealed that under the raw grad $\times$ input target the head collapsed to a near-global importance vector (between/within-sample variance ratio 0.055, top-10 feature overlap 0.791 across samples). A minimum-intervention iteration (v2, "demeaned target") subtracts the batch mean from the supervision signal and restores per-sample discrimination without disturbing the primary task (variance ratio $0.055 \to 0.719$, top-10 overlap $0.791 \to 0.281$, primary AUC unchanged within noise, Section 4.9.4). A further iteration (v3, "primary-task-logit target") replaced the attribution supervision signal with the model's own logit and collapsed the head back below the v1 baseline (variance ratio $0.719 \to 0.043$), documenting that the demeaned target is narrowly effective rather than robustly so (Section 4.13). Audit-log integration has since been implemented — the per-sample attribution vector feeds `AuditLogger.log_attribution` in the companion serving stack (Paper 2 v2) — while cross-dataset reproduction remains deferred.

  Causal Guardrail (CG), a per-prediction reliability flag derived from the causal expert's latent $\boldsymbol{z}$. A W-reconstruction formulation (CG v1) fails at chance-level discrimination because the learned $\boldsymbol{W}$ is too small; a z-space Mahalanobis formulation (CG v2) detects three types of synthetic OOD at 100% TPR with 5% FPR (Section 4.10). A follow-up W-amplification experiment (W init $0.1 \to 0.3$, $\lambda_{\text{recon}} 0.5 \to 2.0$) grows the adjacency matrix $14\times$ in Frobenius norm and from 8.5% to 59.5% active edges with zero primary-

[1]"12 tasks" and "7 experts" here refer to the synthetic v14 benchmark. The operational deployment uses a separate task naming scheme with an active-task count that shifts with label availability (11 as of the July 2026 run), and runs a 6-expert variant that absorbs LightGCN's signal into the unified HGCN input — see Paper 1′s preliminary operational validation section.

[2]DOI: 10.5281/zenodo.19621884

[3]DOI: 10.5281/zenodo.19622052

task cost, partially fixing CG v1 but leaving latent-space CG v2 dominant (Section 4.11). Finding 11′s settled result: the "decorative DAG" from Finding 8 is a training-choice artefact, not an architectural constraint.

- A Counterfactual Probe (CCP, Finding 12) that tests Pearl Rung 3 directly: under $\mathrm{do}(z_j = v)$ interventions the amplified DAG carries a median of 32% and a 95th percentile of 61% of the counterfactual effect, versus 0.16% on the baseline teacher — a 200 × jump that we read as a **preliminary viability signal** for Rung 3 on the amplified teacher, reported as future work rather than a validated result. On baseline checkpoints the DAG is numerically incapable of supporting Rung 3 claims (Section 4.12).
- A CEH v3 variant (Finding 13, **honest negative result**) that replaces the v2 demeaned grad × input supervision with the model's own primary-task logit gradient as target; the head re-collapses to a global importance pattern (variance ratio $0.719 \to 0.043$), demonstrating that the v1→v2 improvement is narrowly target-dependent rather than a robust design principle and closing that branch of CEH exploration (Section 4.13).
- A re-analysis of three successive generations of distillation fidelity reports (Finding 14), all scored on **identical** teacher–student predictions, isolating a behavioural-similarity floor inherent to the cross-architecture PLE → LightGBM pair (teacher–student agreement 0.75–0.82, ranking correlation 0.78–0.91 across all seven distilled tasks) beneath cleanly passing operational metrics (AUC gap $\leq 0.0125$, student calibration error $\leq 0.0114$), and the control-design conclusion that fidelity gates must be partitioned into a blocking operational tier and an informational behavioural tier — because a gate that always fails is a gate that gets bypassed (Section 4.14).

The system, data generator, and ablation scripts are publicly available.[4]

# 2 Related Work

## 2.1 Multi-Task Learning in Recommendations

Shared-bottom networks [1] suffer from negative transfer when tasks conflict. MMoE [2] mitigates this with per-task gating over a shared expert pool, and PLE [3] further separates shared and task-specific extraction layers. AdaTT [4] adds adaptive task-to-task transfer strength.

A common thread: all evaluations use 2–4 tasks of the **same type**. MMoE's Census experiment uses 2 binary tasks. PLE's production evaluation at Tencent uses 2 tasks (CTR + VCR). AdaTT's Alibaba experiment uses 3 closely related engagement tasks. No published PLE/MMoE study evaluates on a mix of binary, multiclass, and regression tasks at the scale we report.

## 2.2 Loss Balancing in MTL

Kendall et al. [6] introduced homoscedastic uncertainty weighting, learning per-task precision parameters that automatically balance loss scales. GradNorm [7] dynamically adjusts loss weights based on gradient norms. MGDA [8] solves a multi-objective optimization problem at each step.

These methods are designed for and tested on scenarios where all tasks share similar loss magnitudes. When binary cross-entropy (scale 0.5), multiclass cross-entropy (scale 3.9 for 50-class), and regression MSE (scale 0.01–1.0) coexist, the implicit assumptions of these balancing methods deserve re-examination.

## 2.3 Gate Design: Softmax vs. Sigmoid

Standard PLE and MMoE use **softmax** gates, enforcing competitive, sum-to-one expert selection. Nguyen et al. [9] demonstrated that sigmoid gating — allowing each expert to contribute independently without competition — achieves higher sample efficiency by eliminating inter-expert competition.

This finding, however, was established on homogeneous task sets. We show that the sigmoid advantage **reverses** when tasks are heterogeneous, because independent expert activation allows high-gradient binary tasks to contaminate experts that multiclass tasks depend on.

# 3 Architecture

## 3.1 PLE with Heterogeneous Expert Basket

Our PLE implementation follows Tang et al. [3] with a key modification: instead of $K$ identical MLP experts, we employ 7 **architecturally distinct** experts, each encoding a different inductive bias:

[4]https://github.com/bluethestyle/aws_ple_for_financial

| Expert | Architecture | Input Dim |
|---|---|---|
| DeepFM | Factorization Machine + DNN | 977D |
| Temporal | Mamba + LNN + Transformer | 116D |
| HGCN | Hyperbolic GCN (Poincaré) | 58D |
| PersLay | Topological (TDA) | 32D |
| Causal | NOTEARS DAG | 129D |
| LightGCN | Graph Convolution | 955D |
| Optimal Transport | Sinkhorn matching | 95D |

Table 1: Expert basket. Each expert receives a different feature subset via FeatureRouter. Total input space: 1211D (Phase 0 v3/v4, 17 groups).

A **FeatureRouter** assigns each expert its designated feature groups, declared in YAML configuration rather than hardcoded. The sum of per-expert input dimensions (2419D) exceeds the total feature space (1211D) because several feature groups are shared across experts with complementary inductive biases.

## 3.2 CGC Gating

The Customized Gate Control (CGC) module computes an attention over the $K$ experts for each task $t$:

$$g_t = \mathrm{softmax}(W_t \cdot h_t) \in \mathbb{R}^K \tag{1}$$

or

$$g_t = \sigma(w_t \cdot h_t) / \sum_j \sigma(w_j \cdot h_j) \tag{2}$$

where $h_t$ is the shared representation. **Softmax** gates enforce competitive, sum-to-one allocation. **Sigmoid** gates allow independent evaluation, normalized post-hoc. The choice between these two is a central finding of this paper (Section 4.2).

## 3.3 Uncertainty Weighting

Following Kendall et al. [6], we learn a log-variance parameter $\log \sigma_t^2$ per task. The weighted loss becomes:

$$\mathcal{L}_t^{\mathrm{uw}} = w_t \cdot \left( \frac{1}{2\sigma_t^2} \cdot \mathcal{L}_t + \frac{1}{2} \log \sigma_t^2 \right) \tag{3}$$

where $w_t$ is the per-task loss weight from configuration. The precision $\frac{1}{2\sigma_t^2}$ and the regularizer $\frac{1}{2} \log \sigma_t^2$ are clamped to stable ranges: $\log \sigma_t^2 \in [-4, 4]$, precision $\in [10^{-3}, 100]$.

The critical detail is that $w_t$ **must multiply the entire expression**, not just $\mathcal{L}_t$. Omitting $w_t$ is the bug we report in Section 4.1.

## 3.4 AdaTT Task Groups

Tasks are organized into 4 Financial DNA groups:

| **Group** | **Tasks** | **Type mix** |
|---|---|---|
| Engagement | next_mcc, top_mcc_shift | 1 mc, 1 bin |
| Lifecycle | churn_signal, product_stability, segment_prediction† | 1 bin, 1 reg, 1 mc† |
| Value | mcc_diversity_trend | 1 reg |
| Consumption | nba_primary, cross_sell_count, will_acquire_* (5 tasks) | 1 mc, 1 reg, 5 bin |

Table 2: Financial DNA task grouping for AdaTT transfer, per `task_groups` in `configs/santander/pipeline.yaml`. The 13-task configuration shown (7 binary, 3 multiclass, 3 regression) is the one used by the v12-era runs (Findings 1–3, 5, and the gate-entropy analyses). A leakage scan in the v13 preprocessing revision removed `segment_prediction` (†) from the live config; the v13/v14 runs (Findings 4, 6–7, and the Finding 14 distillation round) therefore use the remaining 12 tasks (7 binary, 2 multiclass, 3 regression).

AdaTT [4] learns intra-group and inter-group transfer strengths. Note that the Consumption group mixes 5 binary tasks with a multiclass and a regression task — a design that becomes relevant in Section 4.5.

# 4 Results and Analysis

All experiments share the **benchmark_v12** synthetic generation schema: 1M customers; the generation schema defines 13 tasks (7 binary, 3 multiclass, 3 regression). From the v13 preprocessing revision onward, `segment_prediction` is excluded as a leakage risk, so v13/v14 runs train on 12 tasks (7 binary, 2 multiclass, 3 regression). Three version labels recur in this paper and are easily conflated, so we fix their referents here. **benchmark_v12** names the data **generation** schema. **Phase 0 v3/v4** names the **feature-engineering build** version (the 1211-feature / 17-group build described in Section 3). **v13/v14 phase0** name successive **preprocessing revisions** applied when the benchmark moved to the SageMaker pipeline: v13 carries three defects (HMM state duplication, GMM $K = 20$ dead clusters, probability columns passed through the scaler), and v14 fixes them (HMM mode-split, GMM $K = 14$, probability columns excluded from scaling; 1210 features).

Two run families appear in the tables, and every caption is labelled with its source. **Local v12 runs** (Findings 1–3 and 5; RTX 4070, 349 model-input features per the run artefacts) use batch size 5632, learning rate 0.0005, AMP (FP16), cosine annealing with warm restarts ($T_0 = 10$), uncertainty weighting, at 10 or 20 epochs as noted per table. **v14 phase0 SageMaker runs** (Findings 4, 6, 7; ml.g4dn.2xlarge, fp32 DuckDB streaming) run 15 or 30 epochs as noted. Findings 8–13 are diagnostic analyses of local v12 checkpoints plus targeted SageMaker re-runs as noted; Finding 14 re-analyses the v14 distillation round's fidelity reports. All runs reported here are single-seed (42); we make no cross-seed claims (see Limitations). Metrics are reported per task type: Avg AUC (binary), Avg F1-macro (multiclass), Avg MAE (regression), NDCG@3 and Acc@3 (ranking).

## 4.1 Finding 1: The Silent Uncertainty Weighting Bug

When porting Kendall et al.'s uncertainty weighting from our on-premises codebase to the AWS implementation, a subtle bug was introduced. The on-premises code correctly implements Equation 3:

```
# On-premises (correct)
loss = loss_weight * (precision * task_loss +
log_var)
```

The AWS port omitted the per-task `loss_weight` and the clamping:

```
# AWS port (buggy)
loss = precision * task_loss + log_var / 2
```

This omission has two consequences: (1) the `loss_weight` parameter from `pipeline.yaml` — which compensates for cross-entropy scale differences between binary ( 0.5) and 50-class multiclass ( 3.9) tasks — is silently ignored; (2) without clamping, extreme log-variance values can push precision to numerically unstable regions.

The effect is that multiclass tasks, whose raw loss is 8× larger than binary tasks, receive proportionally **less** gradient when precision parameters converge to similar values. Binary tasks, being numerically dominant (7 of 13), further suppress multiclass learning through sheer gradient volume.

| Metric | Δ (fixed − buggy) |
|---|---|
| Avg F1-macro (multiclass) | +0.031 |
| NDCG@3 | +0.018 |

Table 3: Impact of the uncertainty-weighting bug fix (shared-bottom baseline, early benchmark_v12 ablation round). Only the deltas of the two affected metric families were retained in the project's experiment record; the buggy configuration's run artefacts were not preserved, so we do not reproduce absolute values. Binary AUC was essentially unaffected — binary tasks were not the suppressed type. Both deltas exceed any architecture-level change measured in the ablation study.

**Lesson**: Uncertainty weighting implementations must be verified against the **original** mathematical formulation, not just tested for convergence. The buggy version still converges — it simply converges to a suboptimal loss balance that favors the majority task type.

## 4.2 Finding 2: Gate Selection Depends on Task Homogeneity

Existing literature suggests that sigmoid gating outperforms softmax in PLE/MoE architectures [9]. Our results show this holds only when tasks are homogeneous. With 13 heterogeneous tasks, softmax outperforms sigmoid on the ranking metric:

| Variant | Val Loss | Avg AUC | Avg F1m | Avg MAE | NDCG@3 |
|---|---|---|---|---|---|
| Shared Bottom | 14.641 | 0.6691 | 0.2025 | 0.9594 | 0.6849 |
| PLE Softmax | 14.637 | 0.6716 | **0.2026** | **0.9567** | **0.7002** |
| PLE Sigmoid | **14.635** | **0.6724** | 0.2021 | 0.9605 | 0.6943 |

Table 4: Gate type comparison on 13 heterogeneous tasks (benchmark_v12 local runs, 20 epochs $= 2T_0$, seed 42, final epoch). Bold = best per column. Softmax takes the multiclass and ranking columns (F1-macro, NDCG@3); sigmoid edges it on aggregate AUC and val loss by margins several times smaller than the NDCG@3 gap (+0.0008 AUC vs. −0.0059 NDCG@3).

A fourth variant stacking loss-level adaTT on the sigmoid gate is omitted from the 20-epoch table because its 20-epoch run did not complete; at the 10-epoch budget it is a null effect on aggregate AUC relative to plain sigmoid

(0.6728 → 0.6717, $\Delta = -0.0011$) — the result Finding 7′s nine-way comparison builds on.

**Why softmax wins here**: In a softmax gate, each task's attention over experts sums to 1, creating **competitive allocation**. When a binary task claims Expert A with weight 0.6, only 0.4 remains for other experts. This competition **isolates** experts per task type, preventing high-gradient binary tasks from contaminating experts that multiclass tasks depend on.

Sigmoid gates allow each expert to contribute independently. This is beneficial when all tasks have similar gradient magnitudes (homogeneous case), because it enables richer expert combinations. But with 7 binary tasks producing high-gradient signals and 3 multiclass tasks producing relatively smaller gradients, sigmoid allows binary gradients to flow through **all** experts simultaneously, overwhelming the weaker multiclass signal.

The decision criterion is not architectural sophistication but **task-type homogeneity**:

- Homogeneous tasks (all binary, or all regression): prefer sigmoid (richer mixing).
- Heterogeneous tasks (mixed types): prefer softmax (gradient isolation).

## 4.3 Finding 3: Uncertainty Weights Converge Identically Across Architectures

A surprising observation: the learned uncertainty weights at epoch 10 are **virtually identical** between shared-bottom and PLE-softmax:

| Task | Shared-Bottom | PLE Softmax | Δ |
|---|---|---|---|
| will_acquire_payments (binary) | 0.3981 | 0.3974 | −0.0007 |
| nba_primary (7-class) | 0.3353 | 0.3354 | +0.0001 |
| next_mcc (50-class) | 0.3360 | 0.3361 | +0.0001 |
| cross_sell_count (regression) | 0.6652 | 0.6645 | −0.0007 |
| churn_signal (binary) | 0.3432 | 0.3439 | +0.0007 |

Table 5: Learned uncertainty weights at epoch 10 (selected tasks, benchmark_v12 local 10-epoch runs). Differences are at the 4th decimal place regardless of architecture.

This means uncertainty weighting performs **loss-scale normalization** — mapping each task's loss to a comparable magnitude — but does not provide **structural protection** against gradient interference. The structural protection comes from gating (Section 4.2): with near-identical uncertainty weights, softmax still achieves NDCG@3 0.7002 vs.
shared-bottom 0.6849 on the 20-epoch comparison of Table 4, a +0.0153 improvement attributable purely to gate structure.

**Implication**: Practitioners should not rely on uncertainty weighting alone to handle task-type conflicts. It balances scales but does not prevent gradient contamination. Gate design is the actual mechanism for protecting minority task types.

## 4.4 Finding 4: Epoch Budget Sensitivity

A v14 SageMaker matched-pair study (ml.g4dn.2xlarge, identical fp32 DuckDB-streaming pipeline at both budgets) directly tests whether the small AUC gap observed at 10 epochs is an underfitting artefact or a true plateau. The matched setup isolates epoch-budget effect from precision-path drift between the local fp64 baseline and the SageMaker fp32 inference, so the 10ep / 15ep delta in Table 6 is attributable to training duration alone.

| Epoch | Shared-Bottom | PLE Softmax | PLE Full | PLE Full+adaTT |
|---|---|---|---|---|
| 10 | 0.8197 | **0.8233** | 0.8216 | 0.8223 |
| 15 | 0.8015 | **0.8249** | 0.8203 | 0.8213 |
| Δ (15–10) | −0.0182 | +0.0016 | −0.0013 | −0.0010 |

Table 6: Avg AUC at matched 10 / 15 epochs on v14 phase0 (1M rows, ml.g4dn.2xlarge SageMaker spot, fp32 DuckDB stream). PLE variants plateau ($|\Delta| \leq 0.002$) while shared_bottom drops −0.0182 with val_loss climbing 16.58 → 17.64 — the shared representation memorises training noise faster than it generalises.

The matched-pair table makes two corrections to the local-only interpretation in earlier drafts:

**Correction 1: PLE softmax beats shared_bottom at 10 epochs already.** PLE softmax (0.8233) leads shared_bottom (0.8197) by +0.0036 AUC at 10 epochs on identical SageMaker infrastructure, contradicting an earlier local-10-epoch reading where shared_bottom appeared to lead. The local ordering was a precision-path artefact (fp64 pq.read_table vs. fp32 DuckDB), not an epoch-budget deficit of the gated structure.

**Correction 2: shared_bottom overfits, not PLE.** Extending from 10 to 15 epochs degrades shared_bottom by −0.0182 AUC while leaving all three PLE variants flat

($|\Delta| \leq 0.002$). The recommendation to "ensure budget is at least $2 \times T_0$" therefore needs inversion for shared-representation variants: they require **early stopping** once the unfettered shared bottom enters the noise-memorising regime. PLE+CGC's $T \times K$ gate parameters ($T = 13$ tasks, $K = 7$ experts) act as a structural regularizer that absorbs the budget extension without overfitting.

A **PLE-softmax-reg** variant (dropout 0.3, weight_decay $1 \times 10^{-4}$) further improves to AUC 0.8256, val_loss 16.57 (vs. 0.8249 / 17.09 for the unregularised PLE-softmax baseline), showing that even at the plateau the gated variant leaves calibration-loss headroom that conventional regularisation recovers.

**Guideline (revised)**: When comparing MTL structures at the v14 phase0 scale (1M rows, 1210 features, fp32 streaming), the 10-epoch matched comparison is sufficient to discriminate gated from shared architectures. Longer budgets (15+) require regularisation on shared_bottom to prevent overfitting; PLE variants tolerate the extension without it.

## 4.5 Finding 5: GTE Pre-Gating Degrades Mixed-Type Groups

GroupTaskExpert (GTE) adds a pre-gating layer that partitions expert representations by task group **before** PLE gating occurs. The motivation is to strengthen intra-group expert sharing.

However, when a task group contains mixed types — the Consumption group has 5 binary tasks, a multiclass task (`nba_primary`), and a regression task (`cross_sell_count`) — GTE forces shared representation learning across incompatible loss types:

| Variant | NDCG@3 | Avg AUC |
|---|---:|---:|
| PLE Softmax | 0.7002 | 0.6716 |
| PLE Sigmoid + GTE | 0.6632 | 0.6720 |
| $\Delta$ | $-0.0370$ | $+0.0004$ |

Table 7: GTE impact (benchmark_v12 local runs, 20 epochs). NDCG@3 drops substantially when GTE forces mixed-type groups to share pre-gate representations, while aggregate AUC is untouched. The drop is attributable to GTE rather than to the gate type: plain PLE sigmoid scores NDCG@3 0.6943 (Table 4), so GTE costs $-0.0311$ even against its own gate-type baseline.

The mechanism: GTE pools expert outputs at the group level **before** per-task PLE gating can differentiate them. Within the Consumption group, the 5 binary tasks' high-gradient signals dominate the pooled representation, and the group's minority-type tasks — `nba_primary` (the multiclass ranking task behind NDCG@3) and the regression task `cross_sell_count` — receive a representation already biased toward binary decision boundaries. PLE gating, which operates **after** GTE pooling, cannot undo this damage.

**Guideline**: Task groups for GTE (or similar pre-gating mechanisms) must be homogeneous by task type, not by business semantics. If a business-meaningful grouping mixes types, omit GTE and rely on PLE gating alone for inter-expert allocation.

## 4.6 Finding 6: Gate Entropy and Loss–Metric Decoupling

### 4.6.1 CGC Gate Entropy Analysis

To understand **how** PLE's CGC gating allocates experts in practice, we compute the Shannon entropy ratio of each task's gate weight distribution $g_t \in \mathbb{R}^K$ at the end of teacher training (30 epochs). The entropy ratio is defined as:

$$E_t = \frac{H(g_t)}{H_{\max}} = -\frac{\sum_k g_{\{t,k\}} \log g_{\{t,k\}}}{\log} K \tag{4}$$

where $H_{\max} = \log K$ is the maximum entropy for $K$ experts. $E_t = 1$ means perfectly uniform expert utilization; $E_t = 0$ means a single expert captures all weight.

| Task | Layer 1 | Layer 2 | Pattern |
|---|---|---|---|
| top_mcc_shift | 0.347 | — | Single-expert dominance |
| product_stability | 0.431 | — | Single-expert dominance |
| segment_prediction | — | 0.332 | Single-expert dominance |
| cross_sell_count | 0.570 | 0.614 | Moderate diversity |
| churn_signal | 0.691 | 0.860 | Moderate → full diversity |
| nba_primary | 0.851 | 0.839 | Full expert utilization |
| will_acquire_payments | 0.882 | — | Full expert utilization |

Table 8: CGC gate entropy ratios by task and PLE layer (teacher model, 30 epochs — the v12-era 13-task configuration, before `segment_prediction`'s removal). Low entropy ($E_t < 0.45$) indicates 1–2 experts dominate; high entropy ($E_t > 0.80$) indicates all 7 experts contribute meaningfully. This entropy pattern (0.33–0.88) is specific to the synthetic benchmark — Paper 1′s re-measurement on operational real data found a more uniform distribution (0.67–0.97, with few dominant-expert tasks). The no-collapse conclusion itself holds on both.

The entropy ratios reveal three behaviorally distinct task clusters:

**Single-expert dominance** ($E_t$ 0.33–0.43): Tasks such as `top_mcc_shift` (MCC category shift prediction) and `segment_prediction` (4-class customer segment — three named segments plus UNKNOWN) are captured by 1–2 experts. These tasks appear to encode simple patterns that a single specialized expert — DeepFM for transactional features, or HGCN for hierarchical segments — handles near-optimally. The low entropy is not a failure mode; it is efficient routing.

**Moderate diversity** ($E_t$ 0.57–0.72): Tasks such as `cross_sell_count` (count regression) and some binary acquisition tasks draw on 3–4 experts. These tasks likely require both transactional signals (DeepFM) and sequence-level patterns (Temporal), explaining partial expert spread.

**Full expert utilization** ($E_t$ 0.85–0.88): `nba_primary` (7-class next best action) and `will_acquire_payments` (binary) actively use all 7 experts. These tasks encode complex, multi-faceted customer behavior patterns that no single architectural inductive bias fully captures.

#### 4.6.2 Attention Collapse: A Structural Blind Spot

At the attention aggregation level (shared expert aggregation), all 13 tasks exhibit an entropy ratio of **exactly 1.000**. The attention mechanism has not learned to differentiate — it acts as a simple average over the expert pool.

This is structurally significant. The CGC extraction-layer gates (Table Table 8) demonstrate that **the model can learn differentiated expert preferences**, but this capacity is entirely absent at the attention-aggregation level. Two possible explanations:

1. **Gradient starvation**: The attention parameters receive gradients only after passing through per-task tower heads, which already specialize via the extraction-layer gates. By the time signal reaches the attention layer, per-task distinction may be adequately handled upstream.
2. **Parameterization bottleneck**: If the attention query dimension is small relative to the expert embedding space, the attention has insufficient capacity to form task-specific preferences across 7 heterogeneous experts.

In either case, the attention component adds parameters without performing meaningful routing. This is a candidate for architectural simplification — replacing attention aggregation with a fixed average or a learned scalar per task — in future work.

#### 4.6.3 Loss–Metric Decoupling at 30 Epochs

Extending training from 10 to 30 epochs (with $T_0 = 10$, cosine warm restarts) exposes a fundamental tension in composite loss monitoring. The v14 SageMaker re-run (ml.g4dn.2xlarge, fp32 DuckDB-stream pipeline, PLE softmax, seed 42) refines the v13 trajectory by reaching higher absolute AUC magnitudes but preserves the same decoupling pattern.

| Epoch | Val Loss | Avg AUC | NDCG@3 | Avg MAE |
|---|---|---|---|---|
| 1 | 19.71 | 0.7032 | 0.6321 | 0.4322 |
| 5 | 17.51 | 0.7903 | **0.7904** | 0.3717 |
| 10 | 16.44 | 0.8142 | 0.7655 | 0.3382 |
| 13 | **16.20** | 0.8239 | 0.7134 | **0.3291** |
| 15 | 16.38 | **0.8272** | 0.7707 | 0.3216 |
| 20 | 18.43 | 0.8196 | 0.7702 | 0.3102 |
| 21 | 17.35 | 0.8210 | 0.7692 | 0.3203 |
| 25 | 17.92 | 0.8147 | 0.7565 | 0.3199 |
| 30 | 19.04 | 0.8133 | 0.7726 | 0.3166 |

Table 9: Loss–metric decoupling over 30 epochs on v14 phase0 (teacher = PLE softmax, 1M customers, $T_0 = 10$ cosine warm restarts). Val loss reaches its minimum at epoch 13 (16.20) and rises afterwards (overfit onset); Avg AUC peaks at epoch 15 (0.8272) and declines −1.4%p by epoch 30; NDCG@3 peaks remarkably early at epoch 5 (0.7904) and recovers partially by epoch 30 (0.7726, −1.8%p from peak). Avg MAE continues improving through epoch 20 then stabilises. Bold = metric peak.

The v14 trajectory shows three task-type-specific peaks at different epochs:

- **Avg AUC** (binary tasks): peaks at epoch 15 (0.8272), then declines to 0.8133 by epoch 30 (−1.4%p). This is more pronounced than the v13 −0.4%p decline because the cleaner v14 features push the binary ceiling higher early in training, leaving more room for overfitting in the second cosine cycle.
- **NDCG@3** (ranking quality): peaks remarkably early at epoch 5 (0.7904) — well before any cosine restart. The recommendation ranking quality is sensitive to the initial high-LR phase: as the cosine schedule decays the LR through epoch 10, the gate's expert preferences harden in directions that hurt rank-aware NDCG even as AUC continues to improve. Post-epoch-15 NDCG oscillates in 0.71–0.79 with no monotone direction.
- **Avg MAE** (regression tasks): improves steadily 0.4322 → 0.3102 through epoch 20, then plateaus. Composite val_loss shows the reverse U-shape (16.20 minimum at epoch 13) because the binary task cross-entropy dominates as MAE saturates.

The root cause carries over from v13: **task-type dominance in composite loss**. Each task type peaks at a different epoch (NDCG@3 at 5, AUC at 15, MAE at 20+), so any single-metric checkpoint selection sacrifices two of the three task types. v14's wider absolute magnitudes (Δ AUC peak-to-final −1.4%p, Δ NDCG peak-to-final −1.8%p) make the trade-off sharper, not gentler.

#### 4.6.4 Cosine Restart Oscillation Across Task Types

Cosine warm restarts ($T_0 = 10$, $T_{\mathrm{mult}} = 2$) create learning rate spikes at cycle boundaries (epoch 10 → second cycle 11–30 wraps around its own mid-point at epoch 20). NDCG@3 exhibits strong oscillation at these boundaries on v14:

| Epoch (event) | NDCG@3 | Change |
|---|---|---|
| 5 (cycle 1 mid) | **0.7904** | global peak — early high-LR phase |
| 10 (cycle 1 end) | 0.7655 | −2.5%p (drift before restart) |
| 11 (restart 1) | 0.7201 | −4.5%p (sharpest drop) |
| 13 (val_loss min) | 0.7134 | recovery deferred |
| 15 (AUC peak) | 0.7707 | recovers +5.7%p but below epoch 5 |
| 21 (LR mini-bump) | 0.7692 | stable through small perturbation |
| 28 (late cycle 2) | 0.7874 | secondary peak, −0.3%p vs. epoch 5 |
| 30 (final) | 0.7726 | stable below initial peak |

Table 10: NDCG@3 oscillation at cosine restart boundaries on v14. The global peak is at epoch 5 (during the high-LR phase of cycle 1) — **not** at the cycle 1 end as one would expect from the v13 trajectory. The first restart drops NDCG@3 by −4.5%p at epoch 11; recovery to 0.7707 at epoch 15 is partial. Late cycle 2 (epoch 28) approaches the global peak again but does not exceed it.

The v14 pattern is asymmetric across task types. Avg MAE continues to improve through the second cosine cycle (0.4322 → 0.3102 by epoch 20) because regression loss landscapes are smooth and the optimiser returns to low-MAE regions after each LR spike. Binary AUC shows a small dip (−0.4%p) at restart 1 and recovers strongly to a new peak at epoch 15. NDCG@3 suffers the largest disruption (−4.5%p at restart 1) because ranking metrics are sensitive to the relative ordering of scores, and LR restarts temporarily scramble the score scale before the model re-converges. Most importantly, the v14 NDCG@3 **global peak** sits inside the high-LR phase of cycle 1 (epoch 5) — well before the AUC peak (epoch 15) — confirming that ranking-aware multiclass tasks have an earlier optimal checkpoint than binary classification on the same composite-loss schedule.

**Implication**: For ranking-sensitive applications, cosine restarts with $T_{\text{mult}} = 1$ (constant cycle length) should be replaced with $T_{\text{mult}} = 2$ (doubling cycle length) or with learning rate warmup-then-decay (no restart), evaluated over a 30-epoch budget.

#### 4.6.5 Checkpoint Selection Criterion

These findings jointly establish that **val loss is an invalid checkpoint criterion** when tasks of different types share a composite loss. The correct approach is a composite checkpoint metric that weights metric semantics by task type:

$$\text{score}_{\text{ckpt}} = \alpha \cdot \text{AvgAUC} + \beta \cdot \text{NDCG@3} + \gamma \cdot (1 - \text{NormMAE}) \quad (5)$$

where $\alpha, \beta, \gamma$ are set to weight task types equally (e.g., $\alpha = \beta = \gamma = \frac{1}{3}$) rather than proportional to task count. On the v14 trajectory of Table 9 no single epoch is optimal for all three task types — NDCG@3 peaks at epoch 5, val loss bottoms at epoch 13, AUC peaks at epoch 15, and MAE keeps improving through epoch 20. A val-loss monitor would freeze epoch 13, sacrificing both the AUC peak two epochs later and the early ranking optimum; the type-weighted composite makes that trade-off explicit instead of hiding it inside the loss scale.

**Guideline**: When regression tasks are present in a heterogeneous MTL system, (1) define a composite checkpoint metric across task types before training begins, (2) checkpoint every epoch and select post-hoc rather than monitoring val loss, and (3) treat val loss as a diagnostic (indicating regression progress) rather than as the primary stopping criterion.

### 4.7 Finding 7: Two Positive Fusion Recipes on Disjoint Axes, Non-Additive Under Composition

After Finding 2 established that the loss-level `adaTT` variant does not affect aggregate AUC at the 12-task scale ($\Delta = -0.001$, null), eight further mechanisms were evaluated on the same benchmark to map out which forms of fusion augmentation (if any) can extract useful signal beyond CGC's gated selection. The nine-way comparison resolves into three regions on v14 phase0. **Representation-additive fusions** with the exception of M1 complement (loss-level adaTT, AdaTT-sp, ECEB, BRP-MV) inject a residual into the primary representation or propagate residual-error gradients into shared experts; the four degrade aggregate AUC at noise level ($-0.001$ to $-0.003$). **M1 complement** is the v14-specific exception that **raises** AUC ($\Delta = +0.0006$), reversing its v13 ranking ($-0.0053$). **Output-space boosting with gradient isolation** (BRP-detached) lifts NDCG@3 by $+0.025$ but degrades aggregate AUC by $-0.0046$ on v14 — a regression from its v13 **tied** verdict ($-0.0007$), because the cleaner v14 features push the CGC baseline up to 0.8234 (vs v13's 0.6728) and the residual bank that previously broke even now competes with a stronger primary tower. **Training-time load-balancing regularisation** (NEAS — an inverse-gate auxiliary supervision) reproduces its v13 positive-AUC verdict on v14 ($\Delta = +0.0017$, vs v13's $+0.0011$), confirming that the mechanism of preventing expert collapse via aux supervision is robust across phase0 generations. The two non-trivial recipes act on disjoint axes — error correction in output space (BRP-detached) vs. expert-collapse prevention at the gate (NEAS) — yet a ninth experiment stacking them produces a **non-additive** outcome: NEAS's AUC lift vanishes ($\Delta = -0.0026$) while BRP-detached's NDCG@3 advantage also fades (0.7579 vs 0.7871 alone), because the shared experts cannot simultaneously generalise for NEAS and specialise for the primary-supporting optimum that BRP-detached relies on.

#### 4.7.1 Five Mechanisms, One Aggregate-AUC Conclusion

**Loss-level adaTT** (the variant reported in Paper 1) adds a weighted cross-task loss term, $L_i + \lambda \sum_{j \neq i} w(i \to j) L_j$, with transfer weights estimated from gradient cosine similarity.

**AdaTT-sp** [4] adds a native-expert residual: after the CGC gate produces the weighted sum, the mean output of the task's own task-specific experts is re-injected, scaled by a learnable scalar.

**Residual complement (M1)**, introduced in this paper, preserves the primary gated output and adds a complementary weighting $(1 - \text{gate\_weights})$ (clamped and renormalised over the expert axis) applied to the same expert outputs as a residual, scaled by a learnable scalar. The intent is to recover intra-task signal from experts the gate down-weighted, without any cross-task mixing.

**ECEB (Error-Conditioned Expert Bank, MV)**, introduced in this paper, is designed specifically to escape the shared structure of the three above: the residual is derived from the gate's **entropy** rather than from the gate's output. Concretely, the recovery path is a task-agnostic consensus (mean over all expert outputs, no gate weighting), scaled at forward time by the product of a per-task learnable scalar $\sigma(w_t)$ and the normalised gate entropy $\frac{H(g_t)}{\log} N$ (per sample). When the gate is confident (low entropy), recovery collapses toward zero; when it is distributed, recovery activates. By construction ECEB eliminates the "gate-derived residual" factor.

**BRP (Boosting-Residual Path, MV)**, also introduced in this paper, removes the additive-on-representation structure altogether. A per-task residual expert bank takes the last CGC layer's `shared_concat` (gate-bypass feature view) and produces a logit residual matching the primary tower's output shape. The residual is trained by MSE against the primary's **detached** prediction error, i.e. $y - \text{activation}(\text{stop\_grad}(\text{primary}))$ (single-stage boosting). The primary pathway trains on ground truth alone; the combined output $\text{primary} + \sigma(\lambda_t) \cdot \text{residual}$ is used for inference and evaluation only. The primary representation is never touched.

**BRP-detached**, a single-line modification of BRP, feeds `shared_concat.detach()` into the residual bank instead of the raw `shared_concat`. This cuts any residual-MSE gradient from flowing back into the shared experts while leaving the primary pathway, parameter count, and training schedule unchanged. The modification was motivated by the per-task analysis of BRP below.

**NEAS (Neglected-Expert Auxiliary Supervision)** adds a per-task auxiliary head consuming the **inverse-gate-weighted aggregation** of the last CGC layer's expert outputs. The auxiliary target is the primary task label, and the auxiliary loss (scaled by `aux_weight = 0.05`) is added to the total loss during training only; inference does not use the auxiliary head at all. The mechanism explicitly forces neglected experts — those the gate de-emphasises for a given task — to retain predictive representations, mitigating expert collapse. NEAS is structurally independent of all residual mechanisms above; it neither injects a residual nor modifies the primary output.

**NEAS + BRP-detached** stacks the two positive mechanisms, testing additivity. Both are enabled with their standalone settings; no other modification.

Results on the 12-task benchmark are reported as a v14 SageMaker re-run (15 epochs, ml.g4dn.2xlarge, fp32 DuckDB-stream pipeline, seed=42) for all nine scenarios; the original 9-way comparison was conducted on v13 phase0 (HMM duplication, GMM K=20 dead clusters, prob-column scaler applied) and is now fully refreshed on v14 phase0 to align with the paper 1 / paper 3 v14 baselines.

| **Fusion** | **Final AUC** | **F1 macro** | **NDCG@3** | **Δ AUC** |
|---|---|---|---|---|
| Shared-Bottom (no gate) | 0.8015 | **0.1426** | 0.7149 | (reference) |
| CGC gate (sigmoid baseline) | 0.8234 | 0.1404 | 0.7619 | — |
| PLE-full (sigmoid+5 toggles) | 0.8204 | 0.1306 | **0.7871** | −0.0030 |
| PLE-full + adaTT | 0.8202 | 0.1311 | 0.7697 | −0.0032 |
| M1 complement | **0.8240** | 0.1366 | 0.7685 | **+0.0006** |
| ECEB (MV) | 0.8231 | 0.1325 | 0.7625 | −0.0003 |
| BRP (MV) | 0.8216 | 0.1327 | 0.7679 | −0.0018 |
| BRP-detached | 0.8188 | 0.1393 | 0.7871 | −0.0046 (degraded) |
| NEAS | 0.8251 | 0.1325 | 0.7692 | +0.0017 (positive) |
| NEAS + BRP-detached | 0.8208 | 0.1330 | 0.7579 | −0.0026 (non-additive) |

Table 11: Fusion-mechanism comparison with NDCG@3 restored. All rows: v14 SageMaker ml.g4dn.2xlarge, 15 epochs. Δ AUC compares each variant to the CGC sigmoid baseline (row 2; the v14 equivalent of v13′s PLE+CGC reference). Four v14 patterns emerge: **(i)** M1 complement is the only mechanism that **raises** aggregate AUC ($\Delta = +0.0006$) — a v14-specific reversal of v13′s M1 ranking (−0.0053) attributable to the cleaner phase0 features (HMM mode-split, GMM K=14, prob-column scaler-excluded); **(ii)** PLE-full and BRP-detached tie for the highest NDCG@3 (0.7871), with PLE-full reaching this via the GTE + LT + HMM-projector stack while BRP-detached reaches it via residual-only output-space correction; **(iii)** shared_bottom's F1 macro (0.1426) tops the v14 column despite its lowest AUC (0.8015) — the no-gate baseline preserves minority-class signal that the gated variants softmax-redistribute away; **(iv)** NEAS reproduces its v13 positive-AUC verdict on v14 ($\Delta = +0.0017$, vs +0.0011 on v13), but BRP-detached degrades on v14 ($\Delta = -0.0046$, vs −0.0007 tied on v13) because the cleaner v14 features push the CGC baseline higher (0.8234 vs v13′s 0.6728), so the residual bank that previously broke even now competes with a stronger

M1′s best AUC at epoch 1 (pre-training) with monotone decline thereafter indicates that training the learnable recovery weight actively degrades performance — random initialisation is a less harmful operating point than the converged weight.

### 4.7.2 Per-Task Breakdown and the Two Outliers

Aggregate deltas are at noise level ($\leq 0.005$) across all variants, but a per-task breakdown reveals three regimes:

- **Gate-saturated tasks** (top_mcc_shift, mcc_diversity_trend) have low gate entropy (ratio $<$ 0.55) and are insensitive to every recovery mechanism ($|\Delta| \leq 0.003$).
- **Gate-distributed tasks with a strong primary** (churn_signal, will_acquire_lending) have high gate entropy (ratio $>$ 0.82) and show the largest M1 degradation ($-0.020$ and $-0.009$).
- **A single positive outlier** is next_mcc (50 classes, near-random base F1 $\approx 0.01$): all three recovery variants improve it by $+0.005$ to $+0.008$. The gain is large relative to the base but small in absolute terms; we attribute it to the near-floor starting point rather than to a genuine recovery effect.

The remaining 7 tasks fall within noise ($|\Delta| \leq 0.005$).

### 4.7.3 Gate Entropy Correlation: Weak Signal

To test whether gate entropy structurally predicts recovery benefit, we extract per-task gate weights from the final CGC layer of the joint_full checkpoint and correlate task-level entropy with each variant's $\Delta$:

- Loss-level adaTT: $r = -0.31$
- AdaTT-sp: $r = -0.32$
- M1 complement: $r = -0.40$

All three correlations are negative (higher entropy $\rightarrow$ more harm) with consistent sign, but with $n = 13$ and $p \approx 0.18$ none meets conventional significance. The two outliers — churn_signal and next_mcc — are better explained by task-specific factors (label construction for churn_signal, near-floor base rate for next_mcc) than by gate entropy. Gate entropy cannot therefore be claimed as a structural predictor of recovery benefit on this benchmark.

### 4.7.4 BRP, BRP-detached, and NEAS: Diagnosing the Mechanisms

The first four augmented variants (loss-level adaTT, AdaTT-sp, M1, ECEB) all inject a residual additively into the primary representation and produce monotone degradation. BRP is the one variant in the family that places the residual in output space. BRP's aggregate AUC is nevertheless the **lowest** of the five non-baseline runs ($\Delta = -0.0078$), but the drop is accompanied by the **highest** F1 macro and NDCG@3 of those five ($+0.0115$ and $+0.0219$ over CGC). The BRP result therefore looked at first like a task-balance trade-off rather than a success.

A per-task breakdown tells a more specific story. next_mcc (50-class, baseline macro-F1 at near-random 0.0100) improves to 0.0440 (+340% relative) under BRP — the hard-task rescue effect. In the opposite direction, churn_signal — the task with the highest binary AUC in the baseline (0.6868) — drops to 0.6512 ($-0.036$) under BRP, and this single task accounts for most of the aggregate AUC loss. Excluding churn_signal, BRP's binary AUC sits near baseline; the remaining five binary tasks each lose between $-0.001$ and $-0.010$.

The mechanism is a shared-expert gradient leak. BRP's residual bank consumes `shared_concat`, and residual-MSE gradients propagate back into the shared experts — only the residual **target** was detached, not its input. For tasks where the primary is already near its ceiling, the shared experts had converged to a primary-supporting optimum and the additional MSE pressure pulls them off it. For tasks where the primary struggles (next_mcc), the residual supplies signal the primary could not extract. The aggregate drop is therefore not an algorithmic limitation of output-space boosting — it is an implementation artefact of training the residual bank through the shared representation.

BRP-detached tests this directly. Swapping `shared_concat` for `shared_concat.detach()` in the residual bank's input — no parameter change, no training-schedule change — produces the following per-task pattern:

| Task | CGC | BRP | BRP-detached | Verdict |
|---|---|---|---|---|
| churn_signal (AUC) | 0.6868 | **0.6512** | **0.6852** | restored |
| will_acquire_lending (AUC) | 0.6549 | 0.6453 | 0.6553 | restored |
| will_acquire_deposits (AUC) | 0.6534 | 0.6493 | 0.6536 | restored |
| will_acquire_investments (AUC) | 0.6754 | 0.6719 | 0.6764 | restored |
| next_mcc (F1 macro) | 0.0100 | **0.0440** | **0.0356** | retained (+256%) |
| remaining 7 tasks | — | ±0.002 | ±0.002 | unchanged |

Table 12: Per-task comparison of BRP and BRP-detached versus CGC on the subset of tasks where BRP materially changed the metric. Detaching `shared_concat` restores every easy-task AUC loss (−0.036 on churn_signal drops to −0.002) while retaining the majority of the next_mcc rescue effect (+340% relative reduces to +256%, still dominant over CGC's near-random baseline).

The diagnosis is therefore confirmed at the per-task level. The easy- task AUC loss in BRP was caused by residual-MSE gradients reshaping shared experts; detaching the input cuts that channel, shared experts remain on the primary-supporting optimum, and the residual bank uses only its own parameters to learn the hard-task correction.

**NEAS** takes a different path. Rather than a residual, it attaches an auxiliary supervision signal to the **inverse-gate-weighted aggregation** of expert outputs: each task's aux head must predict the primary label using mostly the experts the gate de-emphasised. This creates gradient pressure on every shared expert to remain predictive even when the task-level gate concentrates. The effect is a training- time regulariser against expert collapse. NEAS's trajectory rises monotonically through all 10 epochs, and its per-task lift is spread across 11 of 12 tasks (six of seven binaries improve by +0.0004 to + 0.0029; nba_primary lifts F1 by +0.0056). Unlike BRP, NEAS does not produce a large rescue on any single hard task — next_mcc's F1 moves only from 0.0100 to 0.0107 under NEAS alone, against +0.0256 under BRP-detached — because its mechanism is prevention-of-loss rather than targeted-correction.

The two positive recipes therefore act on **disjoint axes**:

| Dimension | BRP-detached | NEAS |
|---|---|---|
| Where it acts | Output-space residual | Shared-expert gradients via aux loss |
| Training signal | MSE on primary's detached error | Task loss on inverse-gate aggregation |
| Inference overhead | Non-zero (residual expert) | Zero (training-only regulariser) |
| Parameter addition | 0.36M (residual bank) | 0.17M (aux heads) |
| Aggregate Δ AUC (v14) | −0.0046 (degraded) | +0.0017 (positive) |
| Aggregate Δ AUC (v13) | −0.0007 (tied) | +0.0011 (positive) |
| Typical per-task pattern | One big rescue (+256% next_mcc) | Many small lifts (±0.003) |
| Failure mode if stacked | NDCG@3 advantage erased | AUC lift erased |

Table 13: Structural comparison of the two non-trivial fusion recipes identified on this benchmark. NEAS's positive verdict is robust across v13 and v14 phase0 generations; BRP-detached's tied verdict on v13 degrades on v14 because the cleaner v14 features push the CGC baseline higher (0.8234 vs 0.6728), so the residual bank competes with a stronger primary tower. Per-task pattern descriptions and stacking failure modes carry over qualitatively — the per-task analysis below is reported on v13 phase0 where the diagnostic was originally conducted.

#### 4.7.5 Conclusion: Three Structural Regions, Non-Additive Composition

Across the nine runs, three structural regions organise the space of fusion augmentations on top of a heterogeneous-expert PLE with CGC gating:

1. **Representation-additive fusions fail on aggregate AUC.** Five variants — loss-level adaTT, AdaTT-sp, M1 complement, ECEB, and BRP-MV (whose `shared_concat` input propagates residual gradients into shared experts) — all inject or propagate residual-error signal into the primary-representation path. All five degrade AUC, with a monotone relationship between intervention invasiveness and degradation magnitude (−0.001 to −0.008). The residual's definition (gate inverse, own expert, uncertainty-gated consensus, boosting error) does not change this;

what matters is whether the mechanism reaches into the primary-supporting representation.

2. **Output-space boosting with gradient isolation: tied on v13, degraded on v14.** BRP-detached places the residual in output space and trains it as a boosting correction on the primary's detached error, while detaching the `shared_concat` input so residual-MSE gradients never reach the shared experts. On v13 phase0 the result tied CGC on aggregate AUC ($\Delta = -0.0007$; best epoch +0.0008) and lifted F1 macro by +0.007 / NDCG@3 by +0.015, with +256 % relative rescue on the hardest multiclass task. On v14 the verdict shifts: $\Delta$ AUC = −0.0046 (degraded), but NDCG@3 ties PLE-full as the joint best at 0.7871 (+0.025 vs CGC). The cleaner v14 features raise the CGC primary tower's ceiling, narrowing the head-room the residual bank could exploit on v13.
3. **Training-time load-balancing regularisation: positive on v13 and v14.** NEAS attaches a per-task auxiliary head to the inverse-gate- weighted aggregation of expert outputs, trained against the primary label. It raises aggregate AUC by +0.0011 on v13 and +0.0017 on v14 — the only mechanism whose positive-AUC verdict reproduces across phase0 generations. The mechanism is zero-overhead at inference (training-only regulariser) and adds only 0.17M parameters.

**The two positive recipes act on disjoint axes** (error correction in output space vs. expert-collapse prevention at the gate) and are independently reproducible. A ninth experiment stacking them produces a **non-additive** outcome: NEAS's aggregate AUC lift vanishes ($\Delta = -0.0006$), while BRP-detached's hard-task rescue partially survives (next_mcc F1 +0.0250 vs. +0.0256 standalone). The mechanism-level explanation is that NEAS pushes shared experts toward **generalists** (each must predict under inverse-gate reweighting), whereas BRP-detached holds shared experts on the **primary-supporting optimum** via the primary task loss. The shared experts are a finite resource and cannot simultaneously satisfy both pressures.

The practical reading is therefore per-objective rather than stack- everything:

- If aggregate AUC and uniform cross-task robustness matter most, use **NEAS** (zero inference overhead, +0.0011 AUC on v13 / +0.0017 on v14, monotone training trajectory).
- If multiclass-ranking quality matters most (NDCG@K), **BRP-detached** is competitive on both phase0 generations, lifting NDCG@3 by +0.015 on v13 and +0.025 on v14 — but accept the v14 AUC dip ($\Delta = -0.0046$) as a trade-off where the cleaner features have reduced the residual bank's effective head-room.
- Do **not** stack them as-is; doing so erases NEAS's AUC gain without producing the hoped-for additive lift. A mechanism that resolves the shared-expert conflict (e.g., a scheduler that alternates NEAS and BRP-detached pressure across training phases, or a parameter- sharing adapter between the two heads) is a natural follow-up but not studied here.

### 4.8 Finding 8: The Causal Expert's Adjacency Matrix Was a Dead Parameter

During preparatory diagnostics for a follow-up study reinterpreting the role of the causal expert, an unexpected failure surfaced in the baseline architecture itself: the causal expert's learnable adjacency matrix $\boldsymbol{W} \in \mathbb{R}^{32\times 32}$ — the DAG structure the expert is supposed to learn — had collapsed to essentially zero across every checkpoint we inspected. The effect is general, not an artefact of any particular scenario:

| Checkpoint | $\boldsymbol{W}$ Frobenius | Entries $\lvert W\rvert > 0.01$ |
|---|---|---|
| struct_13_ple_sigmoid (CGC baseline) | 0.0001 | 0% |
| struct_13_residual_complement (M1) | 0.0001 | 0% |
| struct_13_eceb (MV) | 0.0003 | 0% |
| struct_13_brp_detached | 0.0001 | 0% |
| upstream on-prem implementation | 0.0001 | 0% |

Table 14: Causal expert's adjacency matrix across five independently trained checkpoints, including two from the upstream on-prem implementation that predates the port. In every case the Frobenius norm of $\boldsymbol{W}$ is below its random init scale, and not a single off-diagonal entry exceeds 0.01 in magnitude. The DAG the expert is supposed to learn does not exist after training.

#### 4.8.1 Root Cause: A Residual That Bypasses the DAG

The expert's forward pass is

$$\boldsymbol{z}_{\text{hat}} = \boldsymbol{z} + \boldsymbol{z}\boldsymbol{W}^2 \tag{6}$$

where $\boldsymbol{z} = \text{feature\_compressor}(\boldsymbol{x})$ and $\boldsymbol{z}_{\text{hat}}$ feeds a downstream `causal_encoder` MLP. The residual term $\boldsymbol{z}$ carries the full latent content regardless of $\boldsymbol{W}$, so the task loss has no structural incentive to push $\boldsymbol{W}$ away from zero.

The NOTEARS acyclicity and sparsity regularisers both penalise $\boldsymbol{W}$ away from being dense, but neither penalises $\boldsymbol{W} = 0$ — in fact the sparsity term is **minimised** there. The global optimum of the combined objective is therefore $\boldsymbol{W} = 0$, and training converges to it reliably.

The learned gradient carries the same problem. Both the task-loss contribution via $\partial \boldsymbol{z}_{\text{hat}} / \partial \boldsymbol{W}$ and the NOTEARS reconstruction gradient (when added) are proportional to $\boldsymbol{W}$ itself:

$$\frac{\partial}{\partial \boldsymbol{W}} \operatorname{trace}\left((\boldsymbol{W} \odot \boldsymbol{W})^k\right) \propto \boldsymbol{W} \tag{7}$$

Any near-zero initialisation produces a near-zero gradient, so $\boldsymbol{W} = 0$ is a **saddle point** that the optimiser cannot escape on its own.

#### 4.8.2 Patch: Reconstruction Loss + Initialisation Rescale

Two changes, both load-bearing:

1. **Reconstruction regulariser.** The original NOTEARS paper minimises $\|\boldsymbol{X} - \boldsymbol{X}\boldsymbol{W}\|_F^2$ — an explicit reconstruction signal. We adopt the compressed-latent analogue as a third term in `get_dag_regularization()`:

$$\mathcal{L}_{\text{recon}} = \operatorname{mean}\left((\boldsymbol{z} - \boldsymbol{z}\boldsymbol{W}^2)^2\right), \quad \lambda_{\text{recon}} = 0.5 \tag{8}$$

   This re-introduces the direct pressure on $\boldsymbol{W}$ that the original paper relies on.

2. **Initialisation rescale.** The initial $\boldsymbol{W} \sim \mathcal{N}(0, 0.01^2)$ was too small: its $\boldsymbol{W}^2$ entries sit at $10^{-4}$, a scale at which the gradient of either task or reconstruction loss is effectively zero. Rescaling the init to $\mathcal{N}(0, 0.1^2)$ keeps $\boldsymbol{W}^2$ on an $O(10^{-2})$ scale (still a small perturbation to the residual path) while carrying enough magnitude to propagate gradient during early training.

Either change alone is insufficient. Reconstruction without init rescale was verified on a 10-epoch SageMaker run and left $\boldsymbol{W} \approx 0$; init rescale without reconstruction would restore the standard NOTEARS pressure but leave the gradient vanishing problem.

#### 4.8.3 Post-Patch Verification

A 10-epoch SageMaker run with both changes produces a non-trivial DAG for the first time on this codebase:

- $\boldsymbol{W}$ Frobenius: **0.338** (init scale was 0.1)
- $|W|$-threshold sparsity at 0.01: 7.3%
- Acyclicity $h(\boldsymbol{W}) = 0$ (valid DAG)
- Mean self-loop strength: 0.000 (diagonal suppressed as intended)
- Top edges by $W_{ij}^2$: `var_23 -> var_13` (0.019), `var_9 -> var_13` (0.009), `var_15 -> var_11` (0.007)

The sparsity ratio is in the target range (paper-recommended 5–15%) and the top edges show a consistent sink (`var_13`), which is the kind of hub structure a latent DAG ought to exhibit.

#### 4.8.4 Aggregate Task Metrics Are Unchanged

Patching the collapse does **not** change downstream task performance. On the same SageMaker softmax-gate 10-epoch run, aggregate metrics are within noise of the pre-patch softmax baseline:

| Run | AUC | F1 macro | NDCG@3 | MAE |
|---|---|---|---|---|
| Pre-patch local (softmax, $\boldsymbol{W} \approx 0$) | 0.6729 | 0.2009 | 0.6814 | 0.9598 |
| Post-patch SageMaker (softmax, $\boldsymbol{W}$ learned) | 0.6719 | 0.2042 | 0.6875 | 0.9597 |
| $\Delta$ | −0.001 | +0.003 | +0.006 | 0 |

Table 15: Task metric change after the W-collapse patch. Differences across AUC, F1 macro, NDCG@3, and MAE are within the noise band observed across the 9-way fusion comparison (Finding 7). The structural bug was real, but its resolution does not, by itself, translate into task improvement.

#### 4.8.5 Implication

The diagnostic finding is structural but the metric finding is null. Two readings:

1. The causal expert has been contributing to prediction primarily through its `causal_encoder` MLP, which is a regular non-linear transform that does not require a meaningful $\boldsymbol{W}$ to fit. Adding a meaningful $\boldsymbol{W}$ changes the latent pathway ($\boldsymbol{z}_{\text{hat}} = \boldsymbol{z} + \boldsymbol{z}\boldsymbol{W}^2$) by a small amount, but the downstream encoder adapts and the ensemble's final prediction looks the same. The DAG is, in the current architecture, **decorative** rather than functionally used by the prediction.
2. For explainability claims that depend on the DAG (attribution, counterfactual probes, reason-code generation), this matters. Pre-patch the DAG was absent — so any such claim built on `get_causal_graph()` was retrieving essentially noise at init scale. Post-patch the

DAG exists and is structured, but is not itself routed into the prediction path.

This motivates a separate structural study (beyond this paper's scope) that redefines the causal expert's role so the learned DAG has a load-bearing route to prediction, not just to the expert's own internal representation. Candidates include an attribution head that consumes the DAG directly, a routing signal from the causal expert to the per-task gate, and a counterfactual probe head. The W- collapse patch reported here is a pre-condition for any of those explorations — without it, the DAG is not there to be routed.

## 4.9 Finding 9: Causal Explainability Head (CEH) — First Axis-3 Attempt

As the first of several Axis-3 candidates that re-wire the causal expert's role (CEH / CG / CTGR / CRCG / CCP), CEH adds a per-sample per-feature attribution vector on top of the causal expert. The motivation is explicit: now that Finding 8 gives the expert a real DAG, the DAG needs to leave the expert's internals and reach a consumer. CEH is the shortest path from "DAG exists" to "DAG has a per-prediction output that an audit log can persist." The consuming audit log lives in the companion serving paper (Paper 2); we treat CEH here as producing that paper's **optional forensic enhancement**, not a capability Paper 2 depends on.

### 4.9.1 Design

A small MLP head maps the expert's 64-dim output back to an input_dim-wide attribution vector, and is trained via MSE to align with the gradient × input saliency baseline of the expert's own scalar output w.r.t. the input:

$$\mathcal{L}_{\text{attr}} = \text{mean}\left(\Big|\text{head}(\text{output}) - \left(\nabla_x \text{ output}.\sum() \odot x\right)\Big\|_9^2\right)$$

Gradient × input is computed with one extra forward pass on a cloned, `requires_grad=True` copy of the input, so the main forward graph is untouched. Cost is ≈ 14% extra compute inside the causal expert (about 1/7 of shared-expert time). Primary prediction path is unchanged; CEH is a training-time regulariser plus an inference-time side output.

Inference: `expert._last_attribution` exposes a per-sample vector of length input_dim. The pipeline-side integration with the audit log (HMAC-signed persistence, SR 11-7 MRM) is delivered in Paper 2 v2 via `AuditLogger.log_attribution`.

### 4.9.2 MV Result (SageMaker teacher_full + CEH, 10 epochs, softmax gate)

| Metric | Pre-patch (W ≈ 0) | Post-patch (no CEH) | Post-patch + CEH |
|---|---|---|---|
| Primary AUC | 0.6729 | 0.6719 | **0.6734** |
| F1 macro | 0.2009 | 0.2042 | 0.1994 |
| NDCG@3 | 0.6814 | 0.6875 | 0.6842 |
| MAE | 0.9598 | 0.9597 | 0.9609 |
| $\boldsymbol{W}$ Frobenius | 0.0001 | 0.338 | **0.366** |
| $\|W\| >$ 0.01 sparse edges | 0% | 7.3% | **8.5%** |
| Attribution head trained | n/a | n/a | **yes** |

Table 16: CEH MV result. Primary AUC is preserved within noise; $\boldsymbol{W}$ Frobenius and sparse-edge share are both slightly higher than the post-patch no-CEH run, suggesting the attribution-head gradient provides an additional structural signal that reinforces the DAG. Attribution head training was verified by post-hoc inspection of the head's weights and biases (non-zero biases starting from zero-init).

The attribution head's layer-3 bias (init = $\mathbf{0}$) moved to $\|\boldsymbol{b}\| = 0.03 \pm 0.04$ per element and layer-0 bias to $0.08 \pm 0.08$, which is straightforward evidence that the MSE alignment loss was contributing gradient to the head.

### 4.9.3 What the MV Does Not Verify

CEH MV confirms that:

1. The head trains without disrupting the primary pathway.
2. The W-collapse patch (Finding 8) remains intact under the extra attribution-training signal — in fact the DAG's Frobenius norm and sparse-edge count both increase marginally.
3. The per-prediction attribution vector exists and can be consumed downstream.

CEH MV does **not** yet verify:

- **Attribution quality.** The head fits the gradient × input target by construction; whether the learned output carries per-sample signal or merely reproduces a global importance pattern required a dedicated post-hoc evaluation. That evaluation (Section 4.9.3) showed a near-global collapse under the raw target and motivated the v2 iteration (Section 4.9.4), which resolved the collapse.

- **Audit-log utility.** The wiring itself has since been delivered in Paper 2 v2 (`AuditLogger.log_attribution`, an HMAC-signed hash-chained per-prediction record); what this paper does not verify is whether the persisted attributions are **useful** to a regulator — that depends on the quality dimension evaluated below.
- **Downstream metric impact.** CEH's small AUC lift is +0.0015 over post-patch softmax, within the noise band of the 9-way comparison. We do not claim it as a significant improvement.

#### 4.9.4 Attribution Quality Evaluation (Post-hoc)

Because the MV alone only shows the head **trains**, we ran a dedicated post-hoc evaluation on 5,000 validation samples to ask whether the trained head produces meaningful per-sample explanations or merely a smoothed global pattern. Four measurements:

| Measurement | Value | Interpretation |
|---|---|---|
| Spearman corr. (CEH vs. grad × input) | mean 0.259, median 0.252 | Partial fit to training target |
| Between-sample / within-sample variance | 0.055 | Attribution varies much more within a sample than across samples |
| Mean top-10 feature overlap across samples | 0.791 | Different samples share ∼ 80% of their top features |
| Stability under input noise ($\sigma = 0.05$) | Spearman 0.985 | Very stable (trivially consistent with near-global output) |

Table 17: CEH attribution quality measurements on the post-patch CEH checkpoint. Taken together, the low between-over-within ratio (0.055) and high top-10 overlap (0.791) indicate the head has largely collapsed to a global importance vector with only small per-sample perturbations — the opposite of what a per-prediction attribution head should produce.

Per-group attribution mass corroborates the flatness. CEH allocates 32.1% to txn_behavior versus 44.7% under grad × input on the same samples, and overweights product_holdings (12.6% vs. 5.5%) and product_hierarchy (11.3% vs. 3.8%). The learned distribution is demonstrably flatter than its own training target.

**Interpretation.** The head fits its target only weakly ($\rho \approx 0.26$) and discards the per-sample component of that target almost entirely. The most likely mechanism: the target itself — grad × input of the causal encoder's summed output — has a large sample-invariant component, and a 64-hidden single-layer MLP can capture that global component with low loss while ignoring the sample-specific residual. The head is not broken; the target is too flat for a thin MLP to be forced into per-sample discrimination.

**What this rules in/out.**

- Rules in: infrastructure path (Finding 9 MV) is correct — a functional DAG feeds an attribution consumer without destabilising primary prediction or the DAG itself.
- Rules out: the current target design (grad × input of causal-encoder output sum) is **not** sufficient to produce a regulator-usable per-sample (local) explanation — a candidate for the local explainability recommended in FSC AI Guideline §4.4. Additional Axis-3 candidates that depend on per-sample attribution quality (CRCG in particular) should not be evaluated against this baseline without target refinement.

**Target refinement candidates.**

1. **Demeaned target:** subtract the batch mean of grad × input before using it as supervision. Forces the head to learn per-sample deviation from the global pattern rather than re-learning the pattern itself. Smallest code change; tests the "target is too flat" hypothesis directly. **Run; see Section 4.9.4 below.**
2. **Primary-prediction target:** replace grad of the causal encoder output with grad of a specific task logit (e.g., churn_signal). Aligns the attribution with what downstream consumers actually ask explanations for, at the cost of per-task heads.
3. **Larger / deeper head:** doubled hidden dim and an extra layer, to test whether the 64-hidden bottleneck is the limiting factor rather than the target.
4. **DAG-path target:** use the learned $\boldsymbol{W}$ to construct feature-to-output influence paths as supervision, shifting from local gradient-based to structural-graph-based attribution.

#### 4.9.5 Iteration v2: Demeaned Target Resolves the Collapse

We ran candidate 1 (demeaned target) as a minimal-intervention test of the "target is too flat" hypothesis. Config flag `ceh.target_mode: "demeaned"` subtracts the per-feature batch mean from the grad × input supervision before MSE; everything else identical to the v1 MV run (same architecture, same hyperparameters, same data, same 10 epochs on SageMaker g4dn.xlarge). The same post-hoc evaluation on 5,000 validation samples produces:

| **Measurement** | **Raw (v1)** | **Demeaned (v2)** | **Change** |
|---|---|---|---|
| Between-sample / within-sample variance | 0.055 | 0.719 | 13 × larger |
| Top-10 feature overlap across samples | 0.791 | 0.281 | 65% smaller |
| Stability under input noise ($\sigma = 0.05$) | Spearman 0.985 | Spearman 0.953 | Still stable |
| Primary AUC (churn_signal) | 0.6866 | 0.6870 | Within noise |

Table 18: CEH attribution quality, v1 (raw grad × input target) vs. v2 (demeaned target). The two discriminative-power measurements move by more than an order of magnitude while primary AUC is unchanged — the head now learns per-sample deviation rather than reproducing a global importance vector, at no cost to the downstream task.

The result confirms the hypothesis from Section 4.9.3: the collapse was a training-target artefact, not a head-capacity or architectural limitation. Minimum intervention (three lines plus a config flag) restores per-sample discrimination.

Per-group attribution mass also rebalances substantially under the demeaned target. txn_behavior drops from 32.1% to 18.3% and gmm_clustering from 28.5% to 21.4%, while product_hierarchy jumps from 11.3% to 30.3% and product_holdings from 12.6% to 21.3%. The demeaned head prefers feature groups that carry per-sample distinguishing signal (product taxonomy) over the globally high-variance groups that dominated the raw head.

**Caveats.**

- Spearman correlation against **raw** grad × input falls from 0.259 to 0.096 between v1 and v2. This is expected and not a regression: the v2 head is supervised against demeaned grad × input, so the raw version is no longer its training target. Quality is measured by the target-independent metrics in Table 18 (variance ratio, top-K overlap, stability).
- Per-sample discrimination is necessary but not sufficient for regulator-usable explanations. Human evaluation on sample cases and alignment with domain expectations remain outside this paper's scope; audit-log integration is implemented in the companion paper's v2 audit infrastructure.
- We report v2 as a single-seed 10-epoch run on Santander. Cross-seed stability and cross-dataset reproduction are not verified here.

**Next iteration directions (not run).** Primary-task-gradient target remains the natural next test: replacing the causal-encoder output sum with a specific task logit aligns the attribution with what downstream consumers actually ask about ("why did this customer receive a high churn score?"). The demeaned infrastructure stays, only the scalar the gradient is taken against changes. Larger head and DAG-path targets remain as lower-priority alternatives.

### 4.9.6 Why CEH First (Not CG / CTGR / CRCG / CCP)

Of the five Axis-3 candidates, CEH has the smallest structural footprint: a single MLP head with an MSE supervision signal, no change to task routing (vs. CTGR), no serving-time path branching (vs. CG), no cross-module wiring (vs. CRCG tying into Paper 2′s reason generator). This lets us confirm the basic premise — that downstream consumers can extract something meaningful from the now- functional DAG — with the cleanest possible test before committing to a heavier redesign. Follow-up candidates are evaluated against CEH's baseline rather than against no-causal-routing-at-all.

## 4.10 Finding 10: Causal Guardrail (CG) — Second Axis-3 Attempt

Where CEH answers "why did the model recommend this?", CG (Causal Guardrail) answers the adjacent question "can we trust this recommendation?" — a per-prediction reliability signal that routes suspicious inputs to a fallback or to human review rather than silently passing them through. SR 11-7 "known limitations" reporting and EU AI Act Art. 9 risk-management both require a runtime mechanism for this. The Korean FSC Financial Sector AI Guidelines [5] (in force June 2026) likewise call for an ex-post emergency stop (financial-stability principle, §5.2) and human-in-the-loop review for high-risk decisions (auxiliary-means principle, §3); CG is a design candidate for such a per-prediction fallback signal, with production wiring left as future work. The natural hook on the causal expert is its DAG structure.

#### 4.10.1 MV Formulation v1 (W-Reconstruction) Fails

The first formulation reused the NOTEARS reconstruction residual from Finding 8. Training optimises $\|\boldsymbol{z} - \boldsymbol{z}\boldsymbol{W}^2\|_F^2$ to keep $\boldsymbol{W}$ out of the zero saddle; at serving time the per-sample version $\|\boldsymbol{z}_i - \boldsymbol{z}_i\boldsymbol{W}^2\frac{\|^2}{\|}\boldsymbol{z}_i\|^2$ is a direct measure of how well the **learned DAG** reconstructs the input's compressed causal state. A well-fit input reconstructs cleanly; an OOD input should not.

We evaluated on the teacher_ceh_demeaned checkpoint with 5,000 validation samples plus three synthetic OOD probes (uniform random in [−3,3], column-permuted, extreme-tail percentiles). All distributions concentrated within a narrow window near 1.0 (in-dist median 0.9995, p99 1.0055; OOD medians 0.9998–0.9972). At the p95 threshold calibrated on the in-distribution set, TPR on OOD probes was 6.8% / 8.1% / 0% — effectively chance-level versus the 5% FPR baseline.

The failure mechanism is the same "decorative DAG" observation from Finding 8: the learned $\boldsymbol{W}$ is small enough ($\|\boldsymbol{W}\|_F \approx 0.36$ on an $8 \times 8$ matrix, so $\boldsymbol{W}^2$ is $O(0.13)$ in Frobenius) that $\boldsymbol{z}\boldsymbol{W}^2$ barely perturbs $\boldsymbol{z}$. The residual ratio $\approx 1.0$ regardless of whether the input fits the DAG, so the signal is degenerate.

#### 4.10.2 MV Formulation v2 (z-Space Mahalanobis) Works

Since $\boldsymbol{W}$ itself is too weak to drive discrimination, we tested whether the **causal latent $\boldsymbol{z}$** carries distributional signal independently. Let $\boldsymbol{\mu}, \boldsymbol{\sigma}$ be the per-dimension mean and standard deviation of $\boldsymbol{z}$ over the in-distribution batch; the score is the standardised-sum-of-squares $d_i = \sum_j \left(\frac{z_{i,j}-\mu_j}{\sigma_j}\right)^2$.

| Probe | median | p95 | p99 | max |
|---|---|---|---|---|
| In-distribution | 23.6 | 62.8 | 268.1 | 983.6 |
| Uniform random | 749.5 | 1200.9 | 1458.5 | 1936.7 |
| Column-permuted | 537.4 | 1076.0 | 1336.1 | 1873.9 |
| Extreme-tail | 479.3 | 479.3 | 479.3 | 479.3 |

Table 19: z-space Mahalanobis CG v2 score distributions on 5,000 validation samples and three synthetic OOD probes. Every OOD median lies above the in-distribution p99, giving near-perfect separation.

| Probe | TPR @ ID p95 | FPR @ ID p95 |
|---|---|---|
| Uniform random | 100.0% | 5.0% |
| Column-permuted | 100.0% | 5.0% |
| Extreme-tail | 100.0% | 5.0% |

Table 20: OOD detection rates at the recommended CG v2 threshold (in-distribution p95 = 62.8). All three probe types flag at the expected false-positive rate.

CG is therefore operational via the causal latent even though it fails via the DAG weights. The latent carries structure the W-reconstruction does not expose. Serving integration: a caller computes $\boldsymbol{\mu}, \boldsymbol{\sigma}$ once from a reference population, caches them, and at inference time thresholds the per-sample score to decide pass-through vs. fallback.

#### 4.10.3 What CG MV Verifies / Does Not

Verified:
1. The z-space formulation detects three qualitatively different OOD probes at effectively 100% TPR with 5% FPR — the primitive works.
2. v1 vs v2 comparison isolates **where** the signal actually lives: the causal latent, not the learned DAG weights. Reinforces the Finding-8 observation that $\boldsymbol{W}$ is structurally present but under-utilised by the current architecture.
3. Zero training cost: CG is a post-hoc analysis on the existing teacher_ceh_demeaned checkpoint. No new SageMaker job.

Not yet verified:
- **Real-world OOD**: the three probes are synthetic. A production guardrail also needs to fire on realistic distribution drift (temporal shift, subgroup imbalance, adversarial perturbation). These are Paper 2 monitoring responsibilities, not tested here.
- **Downstream metric impact**: CG produces a flag; we have not yet measured whether routing flagged predictions to a fallback improves end-to-end task metrics or calibration. Demands an integration with the 3-layer fallback router.
- **Threshold drift**: $\boldsymbol{\mu}, \boldsymbol{\sigma}$ computed on a reference batch will need periodic recalibration as the input distribution evolves. Not tested here.

#### 4.10.4 Implications for Remaining Axis-3 Candidates

The v1/v2 contrast clarifies what to expect for CTGR and CRCG. Both depend on the **learned causal structure** ($\boldsymbol{W}$) being informative in the current architecture, not merely present. The Finding 10 experiment is direct evidence that $\boldsymbol{W}$ is not currently strong enough for structural uses: CTGR's routing and CRCG's reason paths would inherit the same weak signal that doomed CG v1. Either W needs to be amplified (larger init, stronger recon lambda, or DAG-routed residual path) before those candidates are worth running, or CTGR/CRCG should be redesigned to draw from the latent rather than the weights — mirroring the v1→v2 pivot here. Finding 11 (below) runs the amplification experiment and produces a partially mixed answer.

## 4.11 Finding 11: W-Amplification Test

Finding 10 identified two hypotheses for why CG v1 fails: (a) the learned $\boldsymbol{W}$ is simply too small to drive meaningful reconstruction, or (b) the $\|\boldsymbol{z} - \boldsymbol{z}\boldsymbol{W}^2\|^2$ formulation is structurally limited regardless of $\boldsymbol{W}$'s magnitude. We tested (a) by re-training one scenario (teacher_ceh_w_amp) with two amplification knobs: $\boldsymbol{W}$ init scale $0.1 \rightarrow 0.3$ and $\lambda_{\text{recon}}$ $0.5 \rightarrow 2.0$. All other hyperparameters identical to teacher_ceh_demeaned.

### 4.11.1 Training-Side Result: W Amplifies, Primary Unharmed

| Metric | Baseline | W-amp | Change |
|---|---|---|---|
| $\|\boldsymbol{W}\|_F$ | 0.363 | 5.028 | $\approx 14\times$ |
| Active edges ($\|W\| > 0.01$) | 8.5% | 59.5% | $7.0\times$ |
| Max $\|W_{ij}\|$ | 0.11 | 0.77 | $7.0\times$ |
| Primary AUC (churn_signal) | 0.6870 | 0.6865 | within noise |
| Loss | 25.62 | 25.61 | within noise |

Table 21: W-amplification training-side outcome. The learned adjacency matrix grows by an order of magnitude in every structural measure while primary task metrics are preserved. Finding 8′s "decorative DAG" observation is directly and cheaply reversed.

The amplification is free in task-metric terms: the larger $\boldsymbol{W}$ pushes $\boldsymbol{W}^2$ from Frobenius 0.13 to 2.19, so the SCM residual $\boldsymbol{z} + \boldsymbol{z}\boldsymbol{W}^2$ now contributes a meaningfully non-trivial perturbation, yet primary AUC shifts by only 0.0005 (within noise). The decorative-DAG regime was a training choice (too-small init + too-weak recon term), not an architectural constraint.

### 4.11.2 CG v1 Improves but Remains Architecturally Limited

With a 14× larger $\boldsymbol{W}$, CG v1′s discriminative power does increase but does not approach v2′s ceiling:

| Probe | Baseline TPR @ ID p95 | W-amp TPR @ ID p95 |
|---|---|---|
| Uniform random | 6.8% | 22.7% |
| Column-permuted | 8.1% | 18.6% |
| Extreme-tail | 0.0% | 0.0% |

Table 22: CG v1 OOD detection before and after W-amplification (FPR fixed at 5%). Discrimination improves from chance to weak-but-nonzero for two probe types; extreme-tail stays at zero regardless, exposing a structural limit.

The extreme-tail result is the key signal. That probe sets every sample to the same 99th-percentile vector, so every sample has an identical $\boldsymbol{z}$ and therefore an identical residual — $\|\boldsymbol{z} - \boldsymbol{z}\boldsymbol{W}^2\|$ is a point estimate, unable to discriminate by construction. No amount of W amplification fixes this because the formulation is a **per-sample** measure that offers no vantage point on the **distribution** of inputs. CG v2′s z-space Mahalanobis, by contrast, is explicitly a distance from a reference distribution, so it handles the degenerate-OOD case trivially (100% TPR).

### 4.11.3 CG v2 Unaffected

v2′s discrimination is unchanged after amplification: 100% TPR across the three probes at the recommended p95 threshold. The z-space pathway does not depend on $\boldsymbol{W}$'s magnitude; amplification neither helps nor hurts it.

### 4.11.4 What Finding 11 Settles / Leaves Open

Settled:

1. $\boldsymbol{W}$ is **not** architecturally doomed. With init 0.3 + $\lambda_{\text{recon}} = 2.0$, it grows to a non-trivial sparse structure in 10 epochs, with zero primary-task cost.
2. CG v1′s failure in Finding 10 was **partly** due to small $\boldsymbol{W}$ (now contributes 15–20 TPR points) and **partly** due to a formulation ceiling (extreme-tail probe exposes a structural limit).
3. Latent-space CG (v2) remains dominant for OOD detection; W-based guardrails are a supplement at best, not a replacement.

Open:

- Does amplified $\boldsymbol{W}$ unlock CTGR and CRCG, or do those candidates hit similar structural ceilings? Finding 11 answers only the CG question directly; the training-side amplification means CTGR/CRCG can now at least be **tried** without the Finding 10 preconditions blocking them.
- Does an even more aggressive amplification (init 0.5, $\lambda_{\text{recon}} = 5$, removed sparsity) push v1 to a usable level

or crash primary AUC? Not tested; the $\frac{0.3}{2.0}$ setting was chosen as a single-job minimum-intervention.
- Does latent-space CG's 100% TPR survive **realistic** drift (temporal, subgroup, adversarial) rather than synthetic probes? Paper 2 monitoring scope, not tested here.

#### 4.11.5 Updated Recommendation for Remaining Axis-3 Candidates

Finding 10 recommended either amplifying $\boldsymbol{W}$ or re-designing the siblings around the latent. Finding 11 chooses the first option and shows it partially works: $\boldsymbol{W}$ is now meaningful, but the latent-based formulation still ceiling-beats a W-based one when both are available. The practical recommendation for CTGR / CRCG / CCP is therefore:

1. Run them on a W-amplified teacher, so W is **capable** of carrying structural information.
2. Simultaneously evaluate a latent-based alternative for each candidate. In every case where both are measurable, treat the latent version as the baseline to beat rather than assuming the W-based version is preferable.

### 4.12 Finding 12: Counterfactual Probe (CCP) — Pearl Rung 3 Viability (Preliminary)

The causal expert's original motivation — "A causes B" explanations rather than "A correlates with B" [10] — requires access to Pearl's Rung 3 of the causal hierarchy: counterfactuals, "what would the output have been if feature $j$ had been different?" Finding 9 (CEH attribution) sits on Rung 1 (observation). Finding 10 (CG) is off-ladder (safety monitoring). CCP, in contrast, directly tests whether the learned DAG mediates counterfactual queries.

*This finding is a preliminary, single-seed post-hoc probe on the synthetic benchmark. We report it as a viability signal and a direction for future work, not as a validated Rung-3 result — the numbers below quantify what the amplified DAG carries, not a claim that the system delivers counterfactual explanations in production.*

#### 4.12.1 Formulation

Given a sample with causal latent $\boldsymbol{z} = \text{compressor}(x)$ and the SCM residual $\boldsymbol{z}\boldsymbol{W}^2$, the factual output is $\text{encoder}(\boldsymbol{z} + \boldsymbol{z}\boldsymbol{W}^2)$. An intervention $\text{do}(z_j = v)$ produces $\boldsymbol{z}'$ where $z'_j = v$ and $z'_k = z_k$ for $k \neq j$. We compute three outputs:

- **Factual**: $\text{encoder}(\boldsymbol{z} + \boldsymbol{z}\boldsymbol{W}^2)$.
- **Direct-only**: $\text{encoder}(\boldsymbol{z}' + \boldsymbol{z}\boldsymbol{W}^2)$ — intervened latent direct-fed, but the $\boldsymbol{W}^2$-mediated term is held at its pre-intervention value. This isolates the non-causal ("just change the feature") effect.
- **Full-CF**: $\text{encoder}(\boldsymbol{z}' + \boldsymbol{z}'\boldsymbol{W}^2)$ — intervention propagates through both paths. This is the counterfactual prediction under the learned DAG.

The **mediation ratio** $\|\text{full CF} - \text{direct only}\| / \|\text{full CF} - \text{factual}\|$ measures what fraction of the counterfactual effect is carried by the DAG. Pearl's Rung 3 is viable exactly when this ratio is bounded away from zero.

#### 4.12.2 Evaluation: Pearl Rung 3 Viable Only After W-Amplification

Running the probe over 1,000 validation samples, all 32 causal-latent dimensions, and three intervention values ($v \in \{-2, 0, 2\}$) yields:

| **Metric (median over $32 \times 3$ cells)** | **Baseline (de-meaned)** | **W-amp** | **Ratio** |
|---|---|---|---|
| $\|\boldsymbol{W}\|_F$ | 0.363 | 5.028 | $14\times$ |
| Direct effect $\|\boldsymbol{y}' - \boldsymbol{y}\|$ | 3.66 | 3.45 | similar |
| Total effect $\|\boldsymbol{y}_{\text{cf}} - \boldsymbol{y}\|$ | 3.66 | 3.54 | similar |
| Mediated effect $\|\boldsymbol{y}_{\text{cf}} - \boldsymbol{y}'\|$ | 0.003 | 0.887 | $269\times$ |
| **Mediation ratio** | 0.16% | 32.1% | $200\times$ |

Table 23: CCP evaluation on both checkpoints. At the baseline W scale, mediation is 0.16% of the total counterfactual effect — the DAG carries essentially nothing, matching the decorative- DAG observation from Findings 10–11. With W amplified, mediation rises to 32% at median and 61% at the 95th percentile, indicating that the learned DAG **may** serve counterfactual queries when trained under Finding 11′s amplification recipe — a preliminary signal flagged as future work, not a validated result.

#### 4.12.3 Interpretation

The baseline result is a direct operational consequence of the "decorative DAG" observation: with $\boldsymbol{W}^2$ small, changing one latent value barely perturbs the mediated term $\boldsymbol{z}\boldsymbol{W}^2$, so the counterfactual collapses to "just change one feature in direct-feed" — a degenerate Rung 1 operation, not a Rung 3 query. The amplified checkpoint routes 32% of the effect through the DAG at median; the direct effect still dominates, as partial mediation predicts from Pearl's theory, but Rung 3 is no longer blocked.

Finding 12 therefore sketches the Pearl-ladder picture for the causal expert under this project's Axis-3 candidates (Rung 2 remains unevaluated):

- Rung 1 (observation / association): CEH attribution (Finding 9).
- Off-ladder (safety): CG coherence (Findings 10–11).
- Rung 3 (counterfactuals): CCP mediation via W-amplified DAG (Finding 12, this section).
- Rung 2 (interventions / treatment effect): not yet evaluated. A natural next step would be mini-batch intervention experiments on the teacher (e.g. counterfactual product-offer effects), but this requires causally-interpretable training data beyond the synthetic benchmark used here.

CCP ran entirely as a post-hoc analysis on the existing two checkpoints; the MV cost is zero SageMaker spend on top of Findings 9–11.

## 4.13 Finding 13: CEH v3 Primary-Task-Gradient Target — Negative Result

Finding 9's "not-yet-verified" list closed with a candidate we did not run: replace the attribution target's scalar from the causal encoder's aggregate output to a specific task logit. The motivation was that downstream consumers ask explanations in task terms ("why did this customer receive a high churn score?"), so a target aligned with the task should give a more useful per-sample signal than a target aligned with the expert's internal output.

### 4.13.1 Implementation

A new `target_mode = "primary_task"` on CausalExpert. Target is computed by the PLE training loop rather than the causal expert: an extra forward pass on a grad-enabled clone of the input yields the task logit, its gradient with respect to the input's causal subset is computed with `torch.autograd.grad`, multiplied by the input, and demeaned per column across the batch (matching v2). The attribution head's main-forward output is preserved so the attribution loss integrates cleanly into the main backward pass.

The configuration also keeps W-amplification on (init 0.3, $\lambda_{\text{recon}} = 2.0$) so the learned DAG is structurally meaningful (Finding 11). The specific task is churn_signal, the project's regulator-anchoring binary task.

### 4.13.2 Result: Head Re-Collapses to Near-Global

Running the same post-hoc quality evaluation on 5,000 validation samples:

| Metric | v1 raw | v2 demeaned | v3 primary-task |
|---|---|---|---|
| Spearman (CEH vs. raw grad × input) | 0.259 | 0.096 | −0.035 |
| Between-sample / within-sample variance | 0.055 | 0.719 | 0.043 |
| Top-10 feature overlap across samples | 0.791 | 0.281 | 0.799 |
| Stability under noise ($\sigma = 0.05$) | 0.985 | 0.953 | 0.998 |
| Primary AUC (churn_signal) | 0.6866 | 0.6870 | 0.6873 |

Table 24: CEH v1, v2 (demeaned), and v3 (primary-task-gradient, demeaned) on the same evaluation. v3 trains to a different target without hurting the primary task, but its per-sample discrimination collapses back to v1 levels — top-10 overlap 0.799 and variance ratio 0.043 indicate the attribution head re-learned a global importance pattern rather than sample-specific deviation.

Head parameter norms do grow between v2 and v3 ($\|\text{head.0}\|_F = 3.42 \to 4.54$; $\|\text{head.3}\|_F = 4.15 \to 5.78$), so training signal was flowing; the head is not untrained. Nor is primary task performance sacrificed. The failure is in the target design, not the infrastructure.

### 4.13.3 Why the Task-Logit Target Failed Where the Expert-Output Target Succeeded

A plausible mechanism: the task logit is produced by a final linear layer $\boldsymbol{W}_{\text{task}} \cdot \mathbf{repr} + \boldsymbol{b}_{\text{task}}$. Its gradient with respect to the input is $\boldsymbol{W}_{\text{task}} \cdot \left(\partial \frac{\mathbf{repr}}{\partial} \boldsymbol{x}\right)$. The linear layer's weight $\boldsymbol{W}_{\text{task}}$ is a constant direction; the per-sample variation comes only from the Jacobian of the representation with respect to the input. When this gradient is multiplied by $\boldsymbol{x}$ and demeaned across the batch, the surviving per-sample signal appears to be too small relative to the head's capacity to produce a meaningful per-sample output. The head converges to a flat pattern that best explains the global structure of the demeaned target.

The v2 target, by contrast, is the gradient of the **causal encoder**'s aggregate output — a much wider representational bottleneck with richer per-sample interaction terms. Its demeaned version retains enough sample-specific structure for the head to learn from.

### 4.13.4 What This Rules In / Out

Rules in: the CEH infrastructure (attribution head + target injection + demeaning) is agnostic to target origin; v3 infra works mechanically.

Rules out: **task-logit gradient is not automatically a better CEH target than expert-output gradient.** The demean step that rescued v2 does not generalise to this target. Target design is more sensitive than the v1 →v2 pivot suggested; v2′s success is a specific sweet spot rather than a general recipe.

### 4.13.5 Candidate Next Steps (Not Run)

1. **Task logit without demeaning.** The demean step may be over- subtracting structure specific to task gradients. A non-demeaned task target (with L2 regularisation against global collapse) is the obvious next iteration.
2. **Multi-task gradient aggregation.** $\sum_t w_t \nabla_{\boldsymbol{x}} \text{logit}_t$ with $w_t$ a task-weighting vector. Spreads supervision across multiple task logits to avoid any single logit's global structure dominating.
3. **Integrated Gradients as target.** A path-integrated attribution is known to be more discriminative than single-point grad × input.

Cost: one SageMaker job ($0.13). Negative result treated as an honest closed branch of the target-design search space.

## 4.14 Finding 14: Behavioural-Similarity Floor in Cross-Architecture Distillation, and the Two-Tier Fidelity Gate

The serving stack distils the 12-task PLE teacher into per-task LightGBM students — a **cross-architecture** pair: a smooth mixture-of-experts teacher against a tree ensemble whose decision boundaries are axis-aligned step functions — and validates each student against a fidelity gate before deployment. The v14 distillation round (teacher = the regularised PLE-softmax variant of Finding 4) produced three successive generations of fidelity reports (v3 → v4 → v6 in the run artefacts), all scored on the **same** teacher and student prediction files: the students were never retrained between generations; only the gate's measurement semantics changed. The report triplet is therefore a small controlled dataset about the gate itself, and this section re-analyses it from the measurement side at zero additional training compute. The companion paper treats the governance side of the same episode — audit-trail readability and operator-override risk — in its discussion of fidelity-gate semantics; the two accounts are complementary, not duplicated.

### 4.14.1 Three Gate Generations on Identical Predictions

| Gate | Scope | cal_gap semantics | Verdict |
|---|---|---|---|
| v3 | 10 tasks (incl. 3 fallback-routed) | $\lvert \text{ECE}_{\text{teacher}} - \text{ECE}_{\text{student}} \rvert$ | 0 / 10 pass |
| v4 | 7 distilled tasks | student ECE | 0 / 7 pass |
| v6 | 7 distilled tasks | student ECE | Tier 1: 6 / 7 pass; Tier 2: 7 / 7 flagged |

Table 25: Three generations of the fidelity gate scored on identical v14 teacher–student predictions. Per-task metric values agree across the three reports to floating-point re-evaluation noise; only scope, metric definition, and tier semantics differ. The student fleet never changed — the measurement did.

Three defects were found and fixed in sequence (`core/training/distillation_validator.py`; commits `0d6dc34` and `d7cb38b` in the public repository):

1. **Scope.** The v3 gate scored every task that had a trained student, including three tasks the serving fallback had explicitly routed **away** from soft-label distillation because the teacher was unreliable on them. The gate then penalised those students for not matching the teacher — exactly the divergence the routing was designed to introduce. The starkest case is nba_primary: the fallback student scores F1-macro 0.983 against ground truth versus the teacher's 0.278, and the v3 gate fails it for an "f1_macro_gap" of 0.705. The fix scopes the gate to the `distill_tasks` set (the v4/v6 reports carry an explicit `gate_scope: "distilled_tasks_only"` field).
2. **Calibration semantics.** v3 defined `calibration_gap` as $\lvert \text{ECE(teacher)} - \text{ECE(student)} \rvert$, which fails a well-calibrated student whenever the teacher itself is poorly calibrated (typical for focal-loss training on imbalanced binary tasks). Redefining it as the student's **own** ECE against ground-truth labels collapses churn_signal's value from 0.2452 to 0.0045 — a 54 × change from a definition fix, with no model change.
3. **Tier conflation.** After both fixes the v4 gate still passed $\frac{0}{7}$: every operational metric cleared its threshold on every binary task, but the gate required behavioural-similarity metrics (agreement, ranking correlation) to clear simultaneously. The v6 gate separates the two families into a block-

ing operational tier and an informational diagnostic tier (`informational_failures`, aggregated as `tier2_violation_count`), and the verdict becomes 6 / 7.

#### 4.14.2 The Behavioural-Similarity Floor

Table 26 reports the v6 per-task values for the six distilled binary tasks. The operational metrics clear their thresholds by a wide margin — worst-case AUC gap 0.0125 and worst-case student ECE 0.0114 against thresholds of 0.05 — while the behavioural metrics sit in a narrow band below their thresholds on essentially every task: agreement 0.754–0.820 against 0.85, ranking correlation 0.870–0.913 against 0.90 (only two of six tasks clear it).

| **Task** | **AUC gap** | **Student ECE** | **Agreement** | **Rank corr.** |
|---|---:|---:|---:|---:|
| churn_signal | 0.0033 | 0.0045 | 0.8134 | 0.8949 |
| will_acquire_deposits | 0.0068 | 0.0047 | 0.7545 | 0.8703 |
| will_acquire_investments | 0.0016 | 0.0038 | 0.8186 | 0.8895 |
| will_acquire_accounts | 0.0125 | 0.0114 | 0.7999 | 0.9014 |
| will_acquire_lending | 0.0011 | 0.0038 | 0.8204 | 0.8847 |
| will_acquire_payments | 0.0116 | 0.0076 | 0.7973 | 0.9135 |
| (threshold) | $\leq 0.05$ | $\leq 0.05$ | $\geq 0.85$ | $\geq 0.90$ |

Table 26: Per-task fidelity metrics for the six distilled binary tasks (v6 report; v4 identical, v3 differs on the calibration column by definition — see text). Operational metrics (left pair) pass with 4 ×–10 × headroom; behavioural metrics (right pair) sit at a floor just below their thresholds, uniformly across tasks. The uniformity across tasks and the invariance across three measurement generations mark the floor as a property of the PLE → LightGBM architecture pair, not of any individual student.

The mechanism is the architecture pair itself: the student can match the teacher's global ranking (witness the near-zero AUC gaps) while disagreeing on individual point estimates, because a tree ensemble cannot interpolate the teacher's smooth decision surface. The floor is not a defect to be fixed by retraining — three generations of re-measurement on identical predictions could not move it, and no student-side change would, short of changing the student family.

The seventh distilled task, cross_sell_count (count regression), is the one remaining Tier-1 failure, and it exposes a residual semantics question in the regression branch. Its blocking metric is `rmse_gap` $= 0.3177 > 0.10$ — but the gap is the **absolute difference** between teacher RMSE (1.5245) and student RMSE (1.2068): the student is 0.32 RMSE **better** than its teacher, and the gate blocks it for deviating. The regression branch still scores similarity-to-teacher rather than student-own quality — the same conflation the calibration fix removed from the binary branch. We flag this as an open item rather than a verdict; a deviation bound has legitimate uses (a student that beats its teacher on aggregate RMSE may still have redistributed errors in ways the teacher's validation never exercised), but the asymmetry deserves an explicit decision rather than an inherited default.

#### 4.14.3 From an Always-Failing Gate to Two Tiers

The control-design consequence is the point of this finding. A single-tier gate that requires operational and behavioural metrics to pass together is, for this architecture pair, a gate that **always fails** — and an always-failing gate does not stay in force. The operational path around it (`skip_fidelity_gate=true`) disables the entire control, including the operational checks that were passing and are genuinely load-bearing. Relaxing the behavioural thresholds to the observed floor was rejected for the complementary reason: it erases the metric's remaining value as a drift signal. The adopted design separates verdicts by what each metric measures:

- **Tier 1 (blocking, operational)**: is the production scorer broken? AUC gap and student ECE (binary), F1-macro gap (multiclass), MAE/RMSE gap (regression). Failure blocks deployment.
- **Tier 2 (informational, diagnostic)**: how similar is the student to the teacher? Agreement, ranking correlation, Jensen-Shannon divergence, quartile agreement. Violations are recorded and trended, never blocking — but a sudden collapse (e.g. agreement dropping below 0.5) still surfaces as an anomaly in the `tier2_violation_count` trend.

#### 4.14.4 Position in the Finding Lineage

Finding 14 is the serving-side member of the silent-failure lineage that opened this paper. In Finding 1, a control was silently **inert** while training looked healthy — the loss weights were ignored and every run converged plausibly. Here, a control was loudly **failing** while production quality was healthy. Both are failures of measurement semantics rather than of models, and both were invisible to aggregate dashboards: Finding 1 because nothing alarmed, Finding 14 because everything alarmed. Finding 6′s lesson recurs one level up the stack: just as a composite val-loss is an invalid checkpoint criterion when task types differ in metric semantics, a composite single-tier gate is an invalid deployment criterion when metric families differ in **control** semantics. And Finding 7′s caution that aggregate AUC can mask per-task structure reappears at serving time as operational aggregates masking per-sample behavioural divergence. The division of labour with the companion paper is deliberate: Paper 2 documents what the gate episode means for governance — audit-trail semantics, regulator readability, override risk — while this section documents what it means for measurement: where the floor comes from, which metric definitions were wrong, and what a gate must separate to stay in force.

# 5 Discussion

## 5.1 Practical Guidelines Summary

We distill the fourteen findings into twelve guidelines for practitioners, grouped by the four themes.

### 5.1.1 Loss Dynamics and Gating (Findings 1–6)

1. **Gate selection depends on task-type mix, not on architecture.** Use softmax for heterogeneous task mixes (different loss types); use sigmoid for homogeneous tasks (all binary, or all regression). This is the single highest-impact design decision.

2. **Uncertainty weighting is necessary but not sufficient.** It normalizes loss scales — without it, multiclass tasks are silently suppressed. But it does not prevent gradient contamination. Gate structure provides the actual isolation. Verify the implementation against Equation 3, especially the per-task $w_t$ and clamping.

3. **Do not mix task types in pre-gating groups.** GTE, task-group attention, and similar mechanisms assume intra-group homogeneity. Business-meaningful groups that mix binary and regression tasks will degrade the minority type.

4. **Use a composite checkpoint metric, not val loss.** When regression and classification tasks share a composite loss, regression improvement continuously pulls val loss downward while classification/ranking metrics saturate or regress. Define a type-weighted composite metric (Avg AUC + NDCG@3 + normalized MAE) before training and checkpoint by it (Section 4.6).

5. **Gate entropy reveals architectural waste.** If all tasks show uniform attention-level entropy (ratio = 1.000), the attention aggregation is not performing routing — it is averaging. Audit entropy ratios at both the extraction and attention levels before attributing performance gains to gating mechanisms.

### 5.1.2 Fusion Augmentation (Finding 7)

1. **Match the fusion recipe to the objective; do not stack them.** For aggregate-AUC gains with uniform per-task lift, prefer inverse-gate auxiliary supervision (NEAS). For hard-task rescue at the cost of aggregate AUC parity, prefer output-space boosting with shared-expert gradient isolation (BRP-detached). The two recipes operate on disjoint axes — gate-level load balancing vs. output-level error correction — and stacking them collapses the aggregate-AUC gain because shared experts cannot simultaneously be generalists (NEAS) and primary- supporting specialists (BRP-detached).

### 5.1.3 Causal Expert Reinterpretation (Findings 8–13)

1. **Verify the causal DAG is actually learning before consuming it.** A $\boldsymbol{W}$ that drifts to zero during training is a silent failure mode: NOTEARS acyclicity and sparsity penalties are satisfied trivially at $\boldsymbol{W} = 0$, and an SCM residual forward ($\boldsymbol{z} + \boldsymbol{z}\boldsymbol{W}^2$) still propagates the primary signal through $\boldsymbol{z}$. Monitor Frobenius norm, active-edge count, and acyclicity value $h(\boldsymbol{W})$ at training time, not just final task metrics.

2. **Decorative DAG is a training choice, not an architectural constraint.** With init scale 0.1 and reconstruction-loss weight 0.5, $\boldsymbol{W}$ lands at Frobenius 0.36 on an $8 \times 8$ matrix — "present but too weak to matter". Raising init to 0.3 and $\lambda_{\text{recon}}$ to 2.0 amplifies the learned matrix 14 × with zero primary-task cost. Downstream uses that rely on $\boldsymbol{W}$ carrying structural information should be evaluated against an amplified training run, not the default.

3. **Check attribution-head target design for sample-variance collapse.** A target with large sample-invariant content (e.g., raw gradient × input of an aggregate output) lets a thin MLP re-learn the global pattern and ignore the per-sample residual. Verify that attribution varies sample-to-sample (between/within variance ratio, top-$K$ overlap across samples) before claiming per-prediction explainability. A minimal fix — subtract the batch mean of the target — forces per-sample deviation learning and (in our case) raised the variance ratio from 0.055 to 0.719 at no task cost.

4. **Prefer latent-space over weight-space formulations for guardrails.** A per-sample W-reconstruction residual has no distribution awareness: if every sample produces an identical latent, the residual is a point and the guardrail cannot fire. Mahalanobis-style distance from a cached in-distribution reference

batch is distribution-aware by construction and hit 100% OOD TPR at 5% FPR across three probes in our setup. Default to the latent formulation; use the weight-space version only as a supplement.

5. **Amplify $\boldsymbol{W}$ before relying on causal mediation at inference.** Pearl's Rung 3 counterfactuals — "what would the output have been if feature $j$ had been different?" — require the learned DAG to actually propagate the intervention through $\boldsymbol{W}^2$. At the default training recipe the DAG carries under 1% of the counterfactual effect (Finding 12 baseline); under Finding 11′s amplification recipe the mediation ratio rises to a median of 32% and a 95th-percentile of 61%. Any downstream feature that advertises counterfactual reasoning (regulator-facing explanations, treatment-effect estimates, what-if simulations) must be evaluated on an amplified teacher; evaluating on the default teacher silently reduces the claim to a direct-feed observational probe.

#### 5.1.4 Serving-Side Distillation Control (Finding 14)

1. **Partition fidelity gates by metric semantics, and never leave an always-failing gate in force.** Cross-architecture distillation should be expected to show a behavioural-similarity floor (teacher–student agreement, ranking correlation) beneath cleanly passing operational quality (AUC gap, student calibration), because the student family cannot interpolate the teacher's decision surface. Blocking deployment on the similarity family guarantees permanent failure and invites a blanket override that disables the operational checks too. Block only on student-own quality against ground truth; record similarity as a trended diagnostic. And audit each gate metric's definition for the similarity-vs-quality conflation: a gate that scores $|\mathsf{teacher} - \mathsf{student}|$ where the teacher is not the quality standard will fail students for being better than their teacher.

### 5.2 Limitations

**Synthetic data**: All experiments use a synthetic benchmark with controlled noise profiles. Real production data may exhibit different gradient dynamics due to label sparsity, class imbalance, and non-stationary distributions. We plan to supplement with production results as they become available.

**Single expert basket**: Our findings are specific to PLE with 7 heterogeneous experts. Homogeneous-expert PLE may exhibit different gate dynamics.

**Epoch budget and task-type interaction**: Finding 4 acknowledges that 10-epoch comparisons may be premature. Finding 6 extends this to 30 epochs and confirms that additional epochs help regression but harm classification and ranking. Cross-architecture comparisons (e.g., PLE vs.
shared-bottom) at 30 epochs remain pending.

**Single dataset scale**: While 1M customers is representative of mid-sized financial institutions, the findings may not generalize to internet-scale datasets (100M+ users) where task gradient dynamics differ. Findings 8–13 in particular are reported on a single seed and a single dataset (Santander); cross-seed stability and cross-dataset reproduction are deferred.

**Synthetic OOD probes for CG (Finding 10–11)**: The Causal Guardrail evaluation uses three synthetic out-of-distribution probes (uniform random, column-permuted, extreme-tail). Real-world distribution drift (temporal shift, subgroup imbalance, adversarial perturbation) is expected to differ in both structure and difficulty. CG v2′s 100% TPR at 5% FPR on synthetic probes is a sanity-check ceiling, not a production-ready number.

**Attribution meaningfulness not human-evaluated**: CEH v2 (Finding 9) produces per-sample attributions that discriminate across samples (variance ratio 0.719, top-10 overlap 0.281), but we have not assessed whether the resulting top-$K$ features agree with domain- expert expectations or with alternative attribution methods (Integrated Gradients, DAG-path traversal). A human-evaluation pass is future work.

**Remaining Axis-3 candidates (CTGR, CRCG)**: Findings 10–11 establish the precondition — $\boldsymbol{W}$ is trainable — and CCP has since been evaluated on the amplified teacher (Finding 12). The two remaining candidates, CTGR and CRCG, are not yet evaluated. CG's v1→v2 pivot also introduces a concrete prediction (latent-based formulations should be evaluated alongside weight-based ones), which has not been tested on them.

**Fidelity floor specificity (Finding 14)**: the behavioural-similarity floor (agreement 0.75–0.82, ranking correlation 0.78–0.91 across all seven distilled tasks) is measured for one teacher–student pair (heterogeneous-expert PLE → LightGBM) on one benchmark. The two-tier control principle generalises; the numeric floor does not.

**Deterministic-task re-injection not yet measured**: the 13-task configuration (12 after `segment_prediction`'s later removal) descends from an 18-task draft whose deterministic tasks were removed as label-leakage risks. Whether re-injecting them would contaminate MTL training (gradient domination, gate distortion) is an open experimental question requiring a new training run; no claim is made here.

**Scope boundary — evidential uncertainty head**: Separate from the causal-expert reinterpretation thread above, the implementation ships an `EvidentialLayer` (`core/model/layers/evidential.py`) that emits a per-sample neutral prediction with maximum uncertainty when input features are invalid (NaN/Inf detection) or when a caller-supplied `valid_mask` marks a row as untrustworthy; the layer propagates `valid_mask` forward so downstream losses can exclude invalid rows from gradient updates. This head sits in a complementary role to the Causal Guardrail (CG): CG flags **out-of-distribution** inputs in the causal latent space, whereas the evidential head flags **numerically invalid** inputs at the feature level. We deliberately do not report empirical findings on the evidential head here because it did not emerge from the Axis-3 reinterpretation programme; we mention it only to prevent the reader from conflating its role with CG's.

## 5.3 Relationship to Companion Papers

This paper complements two companion papers from the same project, and its own contribution is best read as a single thread rather than as a catalogue. **Paper 1** (architecture and ablation) establishes the heterogeneous-expert PLE design and validates expert specialization via joint feature+expert ablation; the loss-dynamics and gating results we report in detail (Findings 1–6) are the full empirical account of the ablation that Paper 1 summarises, and where the two coincide we cite Paper 1′s figures rather than re-deriving them. **Paper 2** (serving, reason generation, and regulatory compliance) is self-contained: its audit trail is satisfied by the distilled student's SHAP attributions, the three-agent reason pipeline, and an HMAC-signed hash chain, and its GDPR Art. 22 / EU AI Act Art. 13 argument does not depend on any output of this paper.

Against that backdrop, the thread this paper owns is **what silently fails when multi-task learning scales past the homogeneous-task regime, and the measurement discipline needed to catch it.** The silent-failure lineage runs from an uncertainty-weighting control that is inert while training looks healthy (Finding 1), through gate-entropy and checkpoint pathologies that a composite validation loss hides (Finding 6), to a causal expert whose learnable adjacency matrix silently optimises to zero (Finding 8), to a distillation fidelity gate that fails permanently while production quality is healthy (Finding 14). Each is a failure of measurement semantics rather than of a model, and each is invisible to the metric a practitioner would naturally watch.

Two Axis-3 outputs built on the now-functional DAG — the CEH per-sample attribution vector (Finding 9) and the CG z-space coherence score (Findings 10–11) — can be routed into Paper 2′s audit infrastructure (`AuditLogger.log_attribution` / `AuditLogger.log_guardrail`) as an **optional forensic enhancement**. We are deliberate about the framing: this is an enhancement, not a prerequisite. Paper 2′s regulatory argument stands without it, and the per-prediction causal audit currently runs on the PLE teacher rather than the served student, so emitting these records for production traffic is future work. The counterfactual-probe direction (Finding 12) — testing Pearl Rung 3 on the W-amplified teacher — is likewise left to future work and is reported here only as a viability signal, not a validated result.

Finding 14 has a companion-paper counterpart of a different kind: Paper 2′s discussion of fidelity-gate semantics under cross-architecture distillation documents the governance consequences — audit-trail readability, regulator-facing tier semantics, operator-override risk — of the same defect-and-fix sequence whose measurement side is analysed in Section 4.14. The two accounts share the underlying artefacts (the three fidelity-report generations) but neither duplicates the other's analysis.

# 6 Conclusion

Scaling multi-task learning from 2–4 homogeneous tasks to 13 heterogeneous tasks surfaces four families of dynamics that existing literature, evaluated primarily on homogeneous setups, does not address.

**Loss dynamics and gating** (Findings 1–6): gate type choice — softmax vs.
sigmoid — depends on whether tasks share a loss type, not on architectural preference; uncertainty weighting normalizes scales but does not isolate gradients and converges identically across architectures; pre-gating mechanisms like GTE require type-homogeneous groups; shared-representation baselines overfit extended epoch budgets that gated variants absorb without regularisation; gate entropy analysis shows that extraction-layer gating specializes (entropy ratios 0.33–0.88) while attention-level aggregation collapses to uniform averaging (ratio 1.00); and composite val loss is an unreliable checkpoint signal once regression tasks are present, because their continuous improvement masks classification and ranking degradation.

**Fusion augmentation trade-offs** (Finding 7): a 9-way comparison of representation-level and output-level fusions on top of CGC maps the design space into three regions. Representation-additive fusions — loss-level adaTT, AdaTT-sp, M1 complement, ECEB, and MV BRP — propagate residual-error signal into the primary-representation path and uniformly degrade aggregate AUC, with magnitude scaling in the invasiveness of the intervention. Output-space boosting with shared-expert gradient isolation (BRP-detached) ties CGC on aggregate AUC while lifting F1 macro and NDCG@3 and retaining a +256% relative rescue on the hardest multi-class task. Training-time load-balancing regularisation (NEAS) is the first mechanism of the family to actually raise aggregate AUC ($\Delta = +0.0011$) with near-uniform per-task lifts. The two positive recipes act on disjoint axes and are not additive — stacking them collapses NEAS's AUC gain because the shared experts cannot simultaneously be generalists (NEAS) and primary-supporting specialists (BRP-detached). The practical guidance is per-objective.

**Causal expert reinterpretation** (Findings 8–13): the causal expert's adjacency matrix $\boldsymbol{W}$ collapsed to zero in every trained checkpoint we examined, rendering the expert equivalent to a plain MLP — the same silent-failure pattern as Finding 1, one level down in the architecture. A two-part patch (NOTEARS reconstruction loss + initialisation rescale) restores DAG learning at zero primary-task cost, though the recovered DAG is initially "decorative" — structurally valid and not routed into prediction. On top of the now-functional DAG we build two Axis-3 outputs: a Causal Explainability Head (CEH) producing a per-sample attribution vector (Finding 9 — a first formulation collapsed to a global importance pattern, a minimum-intervention "demeaned target" variant restored per-sample discrimination) and a Causal Guardrail (CG) producing a reliability flag (Findings 10–11 — a W-reconstruction formulation failed at chance-level discrimination, a z-space Mahalanobis formulation reached 100% TPR at 5% FPR on three synthetic OOD probes). A W-amplification experiment establishes that the decorative DAG is a training-choice artefact, not an architectural constraint — init $0.1 \to 0.3$ and $\lambda_{\text{recon}} 0.5 \to 2.0$ grow $\|\boldsymbol{W}\|_F$ 14-fold at zero task cost. These two outputs are offered to the companion serving paper's audit infrastructure as an **optional forensic enhancement** rather than a load-bearing dependency: Paper 2′s regulatory argument is complete without them. A counterfactual probe on the amplified teacher (Finding 12) returns a preliminary Pearl-Rung-3 viability signal — a median 32% mediated counterfactual effect under $\mathrm{do}(z_j = v)$ interventions versus 0.16% on the baseline teacher — which we report as motivation for future work rather than as a validated result, since it rests on a single post-hoc analysis of synthetic interventions.

**Serving-side distillation control** (Finding 14): cross-architecture distillation of the 12-task teacher into per-task gradient-boosted students passes every operational quality check (AUC gap $\leq 0.0125$, student calibration error $\leq 0.0114$) while teacher–student behavioural similarity sits at a structural floor (agreement 0.75–0.82, ranking correlation 0.78–0.91 across all seven distilled tasks) that three successive generations of gate semantics, scored on identical predictions, could not move — because it is a property of the architecture pair, not of any student. A single-tier gate over both metric families fails permanently and invites operator bypass; the adopted two-tier design blocks deployment only on operational defects and downgrades similarity metrics to recorded diagnostics. The lesson closes the loop opened by Finding 1: there, a control was silently inert while training looked healthy; here, a control was loudly failing while production quality was healthy. Both are failures of measurement semantics, not of models.

These findings are not novel algorithms but practical diagnostics and minimum-viable candidates on top of them. We hope they prevent other practitioners from re-discovering the same pitfalls when scaling MTL to real-world heterogeneous task portfolios, and that they motivate later work to validate the positive recipes (NEAS, BRP-detached, CEH v2, CG v2, W-amplified teachers) across additional seeds and datasets.

# 7 Author Contributions

**Seonkyu Jeong** (PM / Lead Architect / Data Scientist): Conceived the study, designed the ablation framework, identified all fourteen findings, wrote the manuscript. Led AI-augmented development methodology.

**Euncheol Sim**: Data pipeline, feature engineering, ablation execution.

**Youngchan Kim**: Model training, mathematical verification, loss weighting implementation.

All authors collaborated through Scrum sprints with rapid feedback cycles.

## 7.1 Funding

This research received no external funding. All costs — including AI development tools, AWS SageMaker cloud training, and operational expenses — were borne by the first author's personal funds. Development was conducted on a single desktop-grade GPU (NVIDIA RTX 4070, 12GB VRAM).

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